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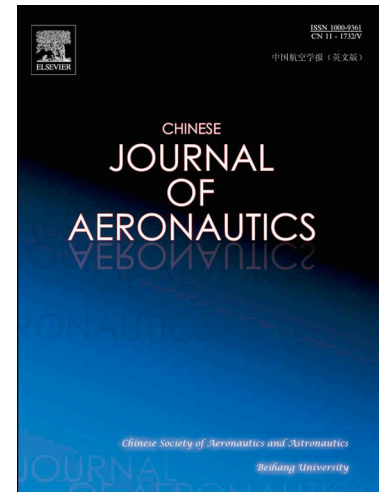
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A novel passive TBD algorithm based on phase correction and observability of moving targets under bistatic observation

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Abstract

A novel Track Before Detect (TBD) algorithm in this work is presented to investigate the target tracking problem under bistatic observation conditions with passive receivers. The proposed algorithm involves deriving Resolution Unit Motion (RUM) model and likelihood of Phase Deviation from Central Mean (PDCM), and utilizing Labeled Multi-Bernoulli (LMB) filter to realize TBD, referred to as the RUM-PDCM-LMB-TBD algorithm. Firstly, a new RUM model for RD (Range Doppler) map is proposed to optimize filtering parameters of LMB-TBD and make more effective use of Doppler information. Additionally, by deriving the phase characteristic distribution of the target, a likelihood function model of PDCM is developed to incorporate the phase information of measurements into the proposed algorithm. Subsequently, to improve the accuracy and stability of target state estimation, these two models, i.e., RUM and PDCM, are combined. Furthermore, via jointly performing TBD and DBT (Detect Before Track) in a single passive radar system, this work elaborates on how to achieve state estimation in Cartesian coordinates; simultaneously, the observability conditions of target states under the bistatic observation model are analyzed in depth. Finally, the effectiveness of the proposed RUM-PDCM-LMB-TBD algorithm is validated through experiments, and the influencing factors of DBT on estimation accuracy are thoroughly analyzed.

Keywords: Target tracking; Doppler effect; Passive filters; Bistatic radar; Track before detect; Labeled random finite set; Observability.

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1. Introduction

In classical radar target tracking, the filter is used to process a set of point measurements from sensors, while the tracking algorithm is used to identify targets of interest and remove clutter based on relationships between measurements across time^{[1]-[3]}. In practice, these point measurements used by the filter are obtained through threshold processing of the raw sensor data, referred to as Detect Before Track (DBT) algorithms. However, for targets in low Signal-to-Noise Ratio (SNR) environments, simple threshold processing cannot provide effective target measurements, leading the filter to form false tracks. An alternative algorithm is known as Track Before Detect (TBD), where the filter processes the raw sensor data directly. For example, the TBD algorithm can be applied to infrared data^[4], and used to deal with range-angle radar resolution cell data^[5] and process sonar data^[6].

With the emergence of the Random Finite Set (RFS) statistical theory, an increasing number of random set filtering solutions for multi-target tracking have been proposed in recent years, such as the Probability Hypothesis Density (PHD) filter based on Poisson random set^[7], Multi-target Multi-Bernoulli (MeMBer) filter based on the multi-Bernoulli random set^[8], and Generalized Labeled Multi-Bernoulli (GLMB) filter based on the labeled random set^[9]. In addition, to further improve the tracking performance of RFS filtering algorithms, several innovative algorithms^{[10]-[12]} have been proposed in our early work. Due to its well-established theoretical framework, the RFS algorithms can effectively handle dynamically varying target numbers with high computational efficiency, demonstrating significant advantages over traditional data association algorithms. Meanwhile, TBD algorithms excel at detecting and tracking weak targets. Therefore, combining RFS and TBD can hold promise for real-time estimation of the number and states of weak targets. Within the RFS framework, various TBD solutions have been proposed, such as the PHD-based MIMO (Multiple-Input Multiple-Output) TBD filter^[13], MeMBer-based grayscale TBD filter^[14], labeled multi-Bernoulli video frame TBD filter^[15], and Bernoulli TBD filter for processing range-azimuth-Doppler measurements in sea clutter scenarios^[16]. However, these filters consider only amplitude information, such as signal strength and grayscale values, without utilizing phase information.

TBD processes raw radar data, and its output depends on the type of radar. Based on the data type, measurements mainly include azimuth, elevation, slant range and Doppler. Some radars can simultaneously acquire multiple types of measurements, making target tracking relatively straightforward. However, for certain radars, the measurement data is insufficient to directly localize targets. For example, some passive radars can only measure angles, while others can only measure distances. Despite these measurement limitations, passive radars offer unparalleled advantages that make them highly valuable for modern surveillance systems. Their primary strength lies in their excellent concealment, as they do not emit any signals, making them undetectable and immune to anti-radiation missiles. Moreover, their operational costs are minimized by exploiting signals from adversarial transmitters. By deploying a synchronized network of low-cost passive receivers, multiple sets of measurements can be obtained, significantly enhancing tracking capability without the expense of dedicated transmitters. Furthermore, passive radar systems exhibit superior resilience in electronic warfare environments, as they are less susceptible to jamming directed against traditional active radar frequencies. These advantages make passive radar have broad application prospects in the military field, leading to extensive research on passive tracking theories and applications^[17]. Generally, passive radars employ multi-sensor joint tracking for target localization, such as TOA (Time of Arrival) localization algorithm in pure range tracking^[18] and AOA (Angle of Arrival) localization algorithm in pure angle tracking^[19].

However, multiple passive sensors cannot always guarantee simultaneous observation of the target. If only a single passive sensor is operational due to factors such as detection range or occlusion, there may be a possibility of losing track of the target. In Ref.[20], a pure angle-only radar under stationary conditions cannot observe the entire state of the target, leading to the problem of solution non-uniqueness. Therefore, observability analysis is required to address such issues and determine the conditions under which the target is observable. For example, in Ref.[21], the observability of a system combining direction-of-arrival and time-difference-of-arrival measurements is analyzed using an information matrix being composed of observation and state transition matrices at multiple time steps. The subsequently established state model is meaningful only if the target's motion state is confirmed to be observable in the model^[22].

Recently, range-based collaborative tracking has emerged as a novel research domain, focusing on joint state estimation using range or TOA measurements from passive radar systems combined with additional information sources. Although Fu et al. implemented single-target tracking using two range-velocity receiving stations, they presumed pre-associated measurements between stations^[23]. Deming et al. developed a TBD algorithm utilizing multiple range-Doppler receivers, employing expectation maximization to estimate target range and radial velocity, but Cartesian state estimation remained unresolved^[24]. Subsequently, Zhou et al. obtained the pseudo-state of the target using a single range-Doppler receiving station in tracking^[25]. If Cartesian coordinates are required, at least two receiving stations are needed for distributed fusion^[26]. These investigations primarily emphasize multi-passive-receiver fusion positioning and consider only scenarios where targets emit signals, neglecting cases where targets merely reflect signals, e.g., the measurement information of the passive receiving station is bistatic information. Consequently, within the RFS framework, this work for the first time investigates both TBD and DBT problems under single passive receiver bistatic observation conditions, and analyzes target state observability in passive receiver systems.

The main contributions of this paper are summarized as follows: (A) A new motion model based on the TBD algorithm is proposed, which can reduce the interference among filtering parameters and better utilize Doppler information. (B) The phase characteristic distribution of the target is derived using the DFT (Discrete Fourier Transform), and boundary conditions are analyzed. Subsequently, the likelihood function of the labeled multi-

Bernoulli TBD algorithm is developed to effectively utilize the phase information of measurements. (C) The observability matrix is derived under the bistatic observation model, and the observability conditions of the target state are analyzed. (D) The distribution of observable degree in different target states is analyzed, and observable degree is used to predict the accuracy of state estimation.

The remainder of this paper is organized as follows. Firstly, Section 2 introduces the random finite set transition density, two models, and measurement computation method involved in passive radar, respectively. Then, Section 3 presents the new motion model based on the TBD algorithm, derives and analyzes the phase characteristics of the target, develops the likelihood function of the filter, and obtains preliminary point estimates. Next, Section 4 establishes a bistatic observation model for point target tracking, and studies the target state observability and observable degree using the observable matrix. Subsequently, Section 5 verifies the feasibility of the new model, demonstrates the distribution of the observable degree of moving targets, and analyzes its impact on estimation accuracy. Finally, Section 6 provides some conclusions.

2. Background

First of all, Section 2.1 introduces the origin and operational process of Bayesian RFS filter, explaining the relationship between the random set transition density and traditional state transition model. Then, Section 2.2 describes the acquisition of range-Doppler data and elaborates on the physical interpretations of bistatic time delay and bistatic Doppler, laying the groundwork for the state estimation modeling in subsequent sections.

2.1. Bayesian filtering with RFS

RFS filtering is a mathematical framework used for estimating the states of multiple targets when the number of targets is unknown and varies over time. Traditional Bayesian filtering techniques are designed for tracking a known and fixed number of targets. However, in many real-world applications, targets may appear or disappear due to births and deaths (e.g., targets entering or leaving the observation area), making the problem inherently random and finite in nature.

RFS filtering treats the collection of target states and observations as random finite sets rather than fixed-dimensional vectors. This approach allows for a unified and rigorous treatment of the uncertainties associated with both the states and number of targets. Given the motion and observation models, the RFS filter uses Bayesian principles to compute the posterior distribution of the state set.

Suppose at time $k-1$ the multi-target posterior is π_{k-1} , and the multi-target measurement set $Z \subset \mathbb{Z}$ is obtained by the sensor. Then, the observation model is given by the likelihood function $g(Z | X)$. Let X_k and Z_k denote the target state set and measurement set at time k , respectively. According to C-K (Chapman-Kolmogorov) equation and Bayes rule, the prediction and update densities can be obtained by

$$\pi_{k|k-1}(X_k | Z_{1:k-1}) = \int f_{k|k-1}(X_k | X_{k-1}) \pi_{k-1}(X_{k-1} | Z_{1:k-1}) \delta X \quad (1)$$

$$\pi_k(X_k | Z_{1:k}) = \frac{g_k(Z_k | X_k) \pi_{k|k-1}(X_k | Z_{1:k-1})}{\int g_k(Z_k | X_{k-1}) \pi_{k|k-1}(X_{k-1} | Z_{1:k-1}) \delta X} \quad (2)$$

The integral here is a set integral, and the set integral is defined by

$$\int f(X) \delta X \triangleq \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\mathbb{X}^n} f(\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}) d(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) \quad (3)$$

The transition density $f_{k|k-1}(X_k | X_{k-1})$ describes the situations of target death, birth, survival and spawning. If birth, death and spawning are not considered, and only survival is accounted for, the state transition model of the target is written as

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}) + \mathbf{v}_{k-1} \quad (4)$$

where $\mathbf{x}_k$ represents an element in the state set, which is also the target state in the traditional non-random set algorithm; $\mathbf{v}_{k-1}$ is the process noise, and all noise in this work is additive noise.

The likelihood function $g_k(Z_k | X_k)$ describes the target's measurement and detection situation under a specific state. If the missed detection component is not considered and only the measurement situation is accounted for, the observation model of the target can be denoted as

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{w}_k \quad (5)$$

where $\mathbf{w}_k$ is the measurement noise. Functions $f(\cdot)$ and $h(\cdot)$ can be referred to as the state transition function and observation function, respectively. If these two functions can be written as

$$f(\mathbf{x}) = \mathbf{F}\mathbf{x} \quad (6)$$

$$h(\mathbf{x}) = \mathbf{H}\mathbf{x} \quad (7)$$

then they can be called the linear motion and observation models, respectively, where $\mathbf{F}$ and $\mathbf{H}$ are the state transition matrix and observation matrix.

2.2. Bistatic time delay and bistatic Doppler frequency

Doppler radar is a type of radar system that measures the velocity of the target by utilizing Doppler effect. The movement of the target causes a change in the frequency of the reflected wave (Doppler shift). By receiving the reflected wave from the radar and analyzing this frequency change, the radial velocity of the target can be measured. If it is an active radar, the radial velocity relative to the receiving station can be obtained. If it is a passive radar, only the bistatic radial velocity can be measured.

The signal path from the transmitter to the receiver consists of two parts: one portion arrives directly at the receiver, while the other is reflected by a target before being received, as shown in Fig. 1.

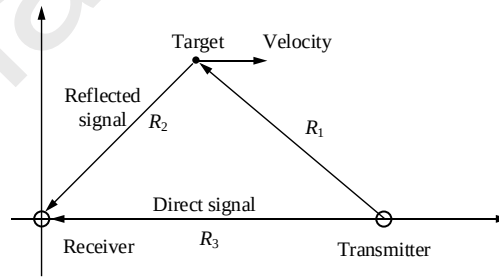


Fig. 1 Signal propagation path.

The bistatic range can be calculated from the bistatic time delay, with the relationship given by

$$\tau_{bi} = \frac{r_{bi}}{c} = \frac{R_1 + R_2 - R_3}{c} \quad (8)$$

where R_1 , R_2 and R_3 denote the transmitter-to-target range, receiver-to-target range and transmitter-to-receiver range, respectively, and c is the speed of light.

Assuming the carrier frequency is f_0 , the relationship between the bistatic Doppler frequency and bistatic velocity is given by

$$f_d = \frac{v_1 + v_2}{c} \cdot f_0 = \frac{1}{\lambda_0} (v_1 + v_2) \quad (9)$$

where λ_0 is the carrier wavelength, v_1 is the radial velocity of the target relative to the transmitting station, v_2 is the radial velocity of the target relative to the receiving station. In Eq. (9), whether v_1 and v_2 have a negative sign depends on the definition of relative velocity. For example, if the velocity of the target approaching the transmitting station is defined as positive, then v_1 does not have a negative sign. Since the measurement model is built upon relative delay and Doppler, it is applicable to scenarios with moving transmitters or receivers.

3. Phase-corrected TBD filter based on unit motion model

The motion and observation models determine the state transition and likelihood functions, respectively, which have a significant impact on the effectiveness of state estimation. Section 3.1 proposes a Resolution Unit Motion (RUM) model. Section 3.2 uses the DFT to derive the phase characteristic distribution of the target on the RD (range-Doppler) map. Finally, Section 3.3 develops a likelihood function of PDCM (Phase Deviation from Central Mean) using phase characteristics and provides the overall pseudocode for the algorithm.

3.1. RUM model

In general, the state vector used by the detection and tracking filter is $[x, v_x, y, v_y]^T$, where x and y represent the positions on the x -axis and y -axis, respectively, while v_x and v_y represent their corresponding velocities. In the TBD domain, the meanings of x and y can vary. For example, in Ref.27, the vector $[x, y]^T$ represents the position of the target in a Cartesian coordinate system. In Ref.28, due to processing raw radar data, $[x, y]^T$ represents the range and Doppler velocity of the target. In Refs.14 and 29, $[x, y]^T$ is derived from image data, so the vector represents pixel coordinates in the image plane.

If $[x, y]^T$ represents Doppler velocity and bistatic distance, respectively, then in the state $[x, v_x, y, v_y]^T$, Doppler velocity and bistatic distance are independent and lack any coupling relationship. This is because, in $[x, v_x, y, v_y]^T$, the quantity associated with x is v_x . Although the RD map shares a similar format to image pixel data, in essence, the distance and Doppler velocity are inherently correlated. For instance, in Ref.28, Doppler velocity is treated as a variance rate of distance. The state vector in this case is reduced to two dimensions (the target's range and Doppler). When applying the CV (Constant Velocity) model, the state transition matrix is expressed as

$$\mathbf{F} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \quad (10)$$

where T is sampling period.

The filter employed in this model takes into account the relationship between distance and Doppler effect. However, due to varying Doppler and time-delay scales across different datasets, the filter cannot accommodate all scenarios, resulting in the lack of a unified standard for parameter settings. For example, the diffusion function used to characterize the target's amplitude characteristics during the TBD process includes several parameters that are dependent on the scales of the horizontal and vertical axes, and these scales cannot be predetermined. If the measurement data have different scale units, these parameters need to be adjusted accordingly. In Ref.14, the diffusion function includes the sensor blur coefficient which is related to the scale intervals of the horizontal and vertical axes. When the actual data scale changes, this parameter requires adjustment. Similarly, in Ref.30, the variance of noise in the likelihood function depends on SNR and sensor blur coefficient. Changes in scale intervals would lead to adjustments in parameters such as the blur coefficient and variance. Since these adjustments cannot be determined in advance, the resulting uncertainty in parameter tuning can significantly impact the stability of the filter.

This section proposes a new nonlinear motion model based on the coupling relationship between echo frequency shift and time delay. By utilizing Doppler and time-delay scales at the current and previous time steps, the model describes the state transition relationship between Doppler units and time-delay units in the TBD scenario. This

approach can effectively eliminate the impact of scale intervals on the sensor blur coefficient, thereby avoiding the need for subsequent parameter adjustments. This resolves the issue of traditional TBD algorithms based on RFS, where Doppler shifts and time delays are processed independently and filtering parameters are inconsistent.

The time delay represents the time elapsed from transmission to reception. The velocity of the signal's movement is the speed of light, so the time delay is only related to distance. From one moment to the next, the change in the bistatic range (the sum of the distances from the target to transmitting station and from the target to receiving station) is calculated as

$$\Delta d = d_k - d_{k-1} = (v_1 + v_2)T = \lambda_0 f_d T = \frac{c}{f_0} \cdot f_d T \quad (11)$$

Thus, the change in latency is

$$\Delta \tau = \tau_k - \tau_{k-1} = \frac{\Delta d}{c} = \frac{f_d}{f_0} \cdot T \quad (12)$$

If f_d being positive indicates that the bistatic range is decreasing, then the latency should also decrease. Therefore, at time k the latency is denoted by

$$\tau_k = \tau_{k-1} - \frac{f_{d,k-1}}{f_0} \cdot T + v_{\tau,k-1} \quad (13)$$

where $v_{\tau,k-1}$ represents process noise.

In this section, the filter involves the state variables of the motion model, expressed as $\mathbf{x} = [x_r, x_d, v_{x_d}]^T$, where x_r and x_d represent the time-delay unit grid index and Doppler unit grid index, respectively, and v_{x_d} corresponds to the target's velocity along the Doppler unit grid, which does not carry direct physical meaning. If the scale parameters are known, the grid indices can be converted to actual values, i.e.,

$$\tau = (x_r - 1)\tau_{\text{gap}} + \tau_{\text{st}} \quad (14)$$

$$f_d = (x_d - 1)f_{d,\text{gap}} + f_{d,\text{st}} \quad (15)$$

where $f_{d,\text{gap}}$ is the interval of Doppler scale, and $f_{d,\text{st}}$ is the first value of Doppler scale. Similarly, τ_{gap} and τ_{st} represent analogous intervals and values for time delay, respectively.

Without considering process noise, substituting Eqs. (14) and (15) into Eq. (13), we can obtain

$$(x_{r,k} - 1)\tau_{\text{gap}} + \tau_{\text{st}} = (x_{r,k-1} - 1)\tau_{\text{gap}} + \tau_{\text{st}} - \frac{(x_{d,k-1} - 1)f_{d,\text{gap}} + f_{d,\text{st}}}{f_0} \cdot T \quad (16)$$

After simplification, the transition relationship between the states $x_{r,k}$ and $x_{r,k-1}$ is calculated by

$$\begin{aligned} x_{r,k} &= x_{r,k-1} - \frac{(x_{d,k-1} - 1)f_{d,\text{gap}} + f_{d,\text{st}}}{f_0 \tau_{\text{gap}}} \cdot T \\ &= x_{r,k-1} - x_{d,k-1} \cdot \frac{f_{d,\text{gap}} T}{f_0 \tau_{\text{gap}}} - \frac{(f_{d,\text{st}} - f_{d,\text{gap}})T}{f_0 \tau_{\text{gap}}} \end{aligned} \quad (17)$$

Next, the state transition for the Doppler unit grid index x_d is modeled. Since v_{x_d} represents the rate of change of x_d , its evolution between time steps $k-1$ and k is given by

$$x_{d,k} = x_{d,k-1} + v_{x_d,k-1}T \quad (18)$$

$$v_{x_d,k} = v_{x_d,k-1} \quad (19)$$

From this, the state transition model corresponding to Eq. (4) can be obtained. When the state variables are Doppler units and time delay units, i.e., $\mathbf{x} = [x_r, x_d, v_{x_d}]^T$, the state transition model is derived as

$$\mathbf{x}_k = \begin{bmatrix} x_{r,k} \\ x_{d,k} \\ v_{x_d,k} \end{bmatrix} + \mathbf{v}_{k-1} = \begin{bmatrix} x_{r,k-1} - x_{d,k-1} \cdot \frac{f_{d,gap}T}{f_0 \tau_{gap}} - \frac{(f_{d,st} - f_{d,gap})T}{f_0 \tau_{gap}} \\ x_{d,k-1} + v_{x_d,k-1}T \\ v_{x_d,k-1} \end{bmatrix} + \mathbf{v}_{k-1} \quad (20)$$

Although the state transition function only involves linear operations, it cannot be written as the form of matrix multiplication in Eq. (6). Therefore, it is a nonlinear motion model.

3.2. Target phase characteristics in distance dimension

In the multi-Bernoulli TBD algorithm based on RFS, updating Bernoulli existence probability requires calculating the likelihood function, which in Ref.14 is defined as

$$g_z(\mathbf{x}_{k|k-1}) \triangleq \prod_m \frac{\varphi_m(\mathbf{z}_m, \mathbf{x}_{k|k-1})}{\phi_m(\mathbf{z}_m)} \quad (21)$$

where the likelihood function $g_z(\cdot)$ is composed of the signal likelihood $\varphi_m(\cdot)$ and noise likelihood $\phi_m(\cdot)$. It can be observed that traditional TBD algorithms only consider the amplitude distribution of measurements without considering the phase information of measurements. If the target phase characteristics are combined to filter out clutter, it can help improve detection performance.

As a representative analytical example, consider a Linear Frequency Modulated (LFM) waveform, whose instantaneous frequency varies linearly with time^[31]. Such a waveform allows closed-form expressions in the RD domain, which facilitates theoretical understanding and parameter sensitivity analysis. It serves as a tractable and analytically convenient model for deriving the theoretical target characteristics.

The radar system receives two signal components: one direct and one reflected. These signals are then mixed and subjected to low-pass filtering. A Fast Fourier Transform (FFT) is applied along the fast-time dimension to obtain range profile data. This process is repeated over multiple pulse repetition intervals, and a second FFT is performed along the slow-time dimension, i.e., the Doppler dimension, to resolve the Doppler frequency of the target. The final output is a two-dimensional matrix, referred to as range-Doppler data or time-delay-Doppler data, in which strong targets appear as distinct peaks. The overall processing flow is illustrated in Fig. 2.

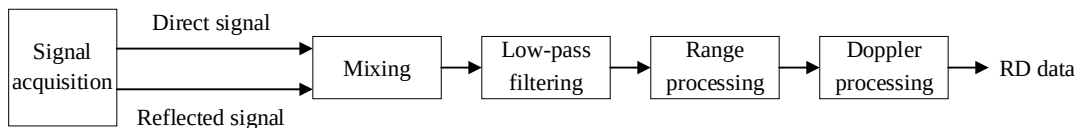


Fig. 2 Processing flow for range-Doppler data acquisition.

Assuming a constant amplitude for the transmitted signal, the LFM signal can be represented as

$$x_T(t) = A \exp \left\{ j2\pi \left(f_0 t + \frac{\mu}{2} t^2 \right) \right\} \quad (22)$$

where f_0 is the carrier frequency, and μ is the modulation slope, which represents the ratio of the modulation bandwidth B to the sweep time T_c , i.e.,

$$\mu = \frac{B}{T_c} \quad (23)$$

The sweep time or sweep period T_c is usually very short. Hence, the target's velocity can be ignored within a single sweep period. The reflected wave phase has a certain time delay relative to the transmitted wave, therefore, the phase functions of the transmitted signal and echo signal with respect to time are, respectively, denoted as

$$\varphi_T(t) = 2\pi \left(f_0 t + \frac{\mu}{2} t^2 \right) \quad (24)$$

$$\varphi_R(t) = 2\pi \left[f_0 (t - \tau) + \frac{\mu}{2} (t - \tau)^2 \right] \quad (25)$$

where the time delay is given by Eq. (8). The receiving station mixes the transmitted signal with received signal. After passing through a low-pass filter to eliminate the sum-frequency component and retain the difference-frequency component, an Intermediate Frequency (IF) signal is obtained. The frequency and phase of the IF signal are both determined by the difference between two signals. Therefore, subtracting the phase functions of the transmitted and received signals yields the phase function of the IF signal, namely,

$$\varphi_{IF}(t) = \varphi_T(t) - \varphi_R(t) = 2\pi\mu\tau t + 2\pi f_0\tau - \pi\mu\tau^2 \quad (26)$$

In a quadrature receiver, the IF signal is obtained in the form of a complex exponential, i.e.,

$$x_{IF}(t) = A \exp \{ j(2\pi f_{IF} t + \phi_{IF}) \} \quad (27)$$

where

$$f_{IF} = \mu\tau \quad (28)$$

$$\phi_{IF} = 2\pi f_0\tau - \pi\mu\tau^2 \approx 2\pi f_0\tau \quad (29)$$

In general, $2\pi f_0\tau$ is much greater than $\pi\mu\tau^2$, so the second term in Eq. (29) can be omitted^[32]. After the signal is converted to digital form, it is stored in a discrete format. Transitioning from continuous to discrete requires sampling. Assuming the sampling rate is F_s , the total number of samples within one sweep cycle is

$$N_s = F_s T_c \quad (30)$$

The sampled signal is written as

$$x[n] = A \exp \left\{ j \frac{2\pi f_{IF} n}{F_s} + j\phi_{IF} \right\} = A \exp \left\{ \frac{jn2\pi f_{IF}}{F_s} \right\} \exp \{ j\phi_{IF} \} \quad (31)$$

Note that the frequency of the IF signal is related to the target distance. To determine the target distance, the frequency of the IF signal must be calculated. When processing discrete signal sequences, DFT is commonly used. However, DFT can only obtain components at specific frequencies. Therefore, to acquire complete frequency component information, a Discrete-Time Fourier Transform (DTFT) analysis is performed first.

Applying DTFT to the signal sequence can yield the continuous frequency distribution, i.e.,

$$\begin{aligned} X(\omega) &= \text{DTFT}\{x[n]\} = \sum_{n=0}^{N_s-1} x[n]e^{-jn\omega} \\ &= A \exp\{j\phi_{\text{IF}}\} P_{N_s}(\omega - \omega_{\text{IF}}) \end{aligned} \quad (32)$$

where

$$\begin{aligned} P_N(\omega) &= \sum_{n=0}^{N-1} e^{-jn\omega} = \frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}} = \frac{e^{j\frac{\omega N}{2}} - e^{-j\frac{\omega N}{2}}}{e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}} \cdot \frac{e^{-j\frac{\omega N}{2}}}{e^{-j\frac{\omega}{2}}} \\ &= \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} e^{-j\frac{\omega(N-1)}{2}} \end{aligned} \quad (33)$$

$$\omega_{\text{IF}} = \frac{2\pi f_{\text{IF}}}{F_s} = \frac{2\pi \mu \tau}{F_s} = \frac{2\pi B \tau}{N_s} = \frac{2\pi B r_{\text{bi}}}{N_s c} \quad (34)$$

Thus, $X(\omega)$ is composed of a real number and an exponential term. For the exponential term, its amplitude is constant at one, and the phase is

$$\varphi_p(\omega) = \phi_{\text{IF}} - \frac{1}{2}(\omega - \omega_{\text{IF}})(N_s - 1) \quad (35)$$

The phase of the exponential term is a linear function. The phase of $X(\omega)$ changes linearly under the condition of ignoring the real coefficient part. The constant gain A can be omitted, and the real part is determined solely by a complex trigonometric function, denoted by

$$Q_N(\omega) = \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} \quad (36)$$

where $Q_N(\omega)$ can be either positive or negative. The phase of a positive number is 0, while the phase of a negative number is π or $-\pi$. Since the phases π and $-\pi$ are equivalent, for simplicity, we can restrict it to π . Therefore, the phase of $X(\omega)$ is represented as

$$\varphi_X(\omega) = \begin{cases} \varphi_p(\omega), & Q_{N_s}(\omega - \omega_{\text{IF}}) \geq 0 \\ \varphi_p(\omega) + \pi, & Q_{N_s}(\omega - \omega_{\text{IF}}) < 0 \end{cases} \quad (37)$$

The amplitude of $X(\omega)$ is denoted as

$$A_X(\omega) = |Q_{N_s}(\omega - \omega_{\text{IF}})| \quad (38)$$

When $\omega = \omega_{IF}$, the amplitude reaches its maximum value, corresponding to the target's position. With 16 sampling points, the amplitude and phase spectrum of $X(\omega)$ within one period are illustrated in Fig. 3.

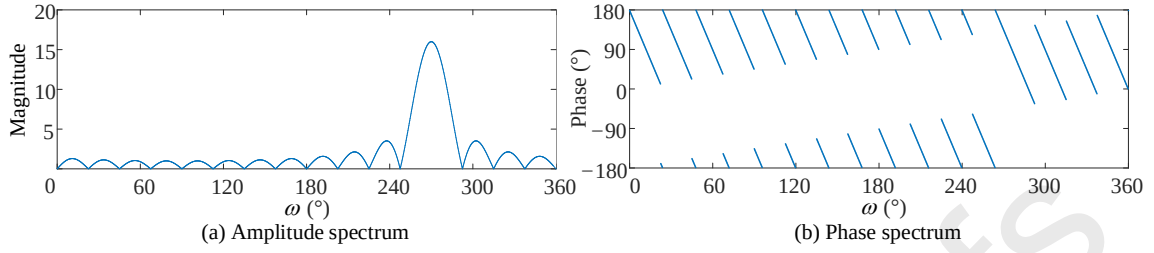


Fig. 3 DTFT of signal. Phase spectrum

In Fig. 3(b), the phase jump occurs due to the change of the positive or negative value of $Q_{N_s}(\omega - \omega_{IF})$, resulting in abrupt phase shifts of π at certain moments, disrupting the original linear phase variation. Whether the phase jumps or not depends on whether the positive or negative value of $Q_{N_s}(\omega - \omega_{IF})$ changes, i.e.,

$$Q_{N_s}(\omega - \omega_{IF}) = \frac{\sin\left[\frac{(\omega - \omega_{IF})N_s}{2}\right]}{\sin\left[\frac{(\omega - \omega_{IF})}{2}\right]} = \frac{\sin\left[\frac{1}{2}\left(\omega N_s - \frac{2\pi Br_{bi}}{N_s c}\right)\right]}{\sin\left[\frac{1}{2}\left(\omega - \frac{2\pi Br_{bi}}{N_s c}\right)\right]} \quad (39)$$

Note that ω_{IF} cannot increase indefinitely. If it exceeds 2π , the peak value cannot be detected. Since ω_{IF} is related to the distance r , the range of values for the distance r is calculated by

$$0 \leq \frac{2\pi Br_{bi}}{N_s c} < 2\pi \Rightarrow 0 \leq r_{bi} < \frac{N_s c}{B} \quad (40)$$

When ω approaches ω_{IF} , we can obtain

$$\lim_{\omega \rightarrow \omega_{IF}} Q_{N_s}(\omega - \omega_{IF}) = \lim_{\omega \rightarrow \omega_{IF}} \frac{\sin\left[\frac{(\omega - \omega_{IF})N_s}{2}\right]}{\sin\left[\frac{(\omega - \omega_{IF})}{2}\right]} = N_s \quad (41)$$

At this point, $Q_{N_s}(\omega - \omega_{IF})$ is a positive number. Therefore, when ω is near ω_{IF} such that $Q_{N_s}(\omega - \omega_{IF})$ is positive or zero, the phase of $X(\omega)$ changes continuously and linearly. There are three cases: both numerator and denominator are positive, numerator and denominator are negative, and the final case occurring at the boundary where the numerator is zero and the denominator is nonzero. First, consider the case where both are positive, i.e.,

$$\begin{cases} 0 < \frac{(\omega - \omega_{IF})N_s}{2} < \pi \\ 0 < \frac{(\omega - \omega_{IF})}{2} < \pi \end{cases} \Rightarrow \begin{cases} \omega_{IF} < \omega < \omega_{IF} + \frac{2\pi}{N_s} \\ \omega_{IF} < \omega < \omega_{IF} + 2\pi \end{cases} \Rightarrow \omega_{IF} < \omega < \omega_{IF} + \frac{2\pi}{N_s} \quad (42)$$

At this boundary, the third case will arise. When an equal sign is taken on the right side of the inequality, the numerator becomes zero while the denominator remains greater than zero, resulting in a phase of zero, which satisfies the condition of the third case. Therefore, the inequality becomes

$$\omega_{\text{IF}} < \omega \leq \omega_{\text{IF}} + \frac{2\pi}{N_s} \quad (43)$$

Next, consider the case where both the numerator and denominator are negative, i.e.,

$$\begin{cases} -\pi < \frac{(\omega - \omega_{\text{IF}})N_s}{2} < 0 \\ -\pi < \frac{(\omega - \omega_{\text{IF}})}{2} < 0 \end{cases} \Rightarrow \begin{cases} \omega_{\text{IF}} - \frac{2\pi}{N_s} < \omega < \omega_{\text{IF}} \\ \omega_{\text{IF}} - 2\pi < \omega < \omega_{\text{IF}} \end{cases} \Rightarrow \omega_{\text{IF}} - \frac{2\pi}{N_s} < \omega < \omega_{\text{IF}} \quad (44)$$

In summary, near ω_{IF} , the phase of $X(\omega)$ remains continuous and linearly varying, equivalent to

$$\omega_{\text{IF}} - \frac{2\pi}{N_s} < \omega \leq \omega_{\text{IF}} + \frac{2\pi}{N_s} \quad (45)$$

The DFT of a discrete signal can obtain information at specific frequencies. Therefore, the DFT can be considered as uniformly-space sampling of DTFT results. For a sequence of length N , performing an N -point DFT is equivalent to applying DTFT to the sequence and then uniformly-space sampling N points within the range $[0, 2\pi)$, with a sampling interval of $2\pi/N$. After performing DFT, the signal is written as

$$\begin{aligned} X[k] &= \text{DFT}\{x[n]\} = X(\omega) \big|_{\omega=2\pi k/N_s} \\ &= A \exp\{j\phi_{\text{IF}}\} P_{N_s} \left(\frac{2\pi}{N_s} k - \omega_{\text{IF}} \right) \end{aligned} \quad (46)$$

where k is an integer that takes values in the range $[0, N_s)$. The phase of $X[k]$ is

$$\varphi_X(k) = \begin{cases} \varphi_p \left(\frac{2\pi}{N_s} k \right), & Q_{N_s} \left(\frac{2\pi}{N_s} k - \omega_{\text{IF}} \right) \geq 0 \\ \varphi_p \left(\frac{2\pi}{N_s} k \right) + \pi, & Q_{N_s} \left(\frac{2\pi}{N_s} k - \omega_{\text{IF}} \right) < 0 \end{cases} \quad (47)$$

and the magnitude of $X[k]$ is

$$A_X(k) = \left| Q_{N_s} \left(\frac{2\pi}{N_s} k - \omega_{\text{IF}} \right) \right| \quad (48)$$

When the length of the sequence signal N is 16, performing an N -point DFT is equivalent to sampling the result of the DTFT at N points within one period. The result is shown in Fig. 4. Each sampling point represents a frequency interval. If an N -point sequence undergoes an N -point DFT, the number of frequency intervals will be N , and the radar resolution unit is also N . It can be observed that the amplitude is non-zero only at certain values, and if the amplitude is zero, the phase is also zero. This is because the sampling point happens to be at the junction of lobes. The result of the DFT discards all frequency components between intervals, even though the N -point DFT uniquely represents the information of the sequence $x[n]$ in the frequency domain. However, it clearly cannot provide sufficient detailed resolution to display the spectral characteristics of $x[n]$.

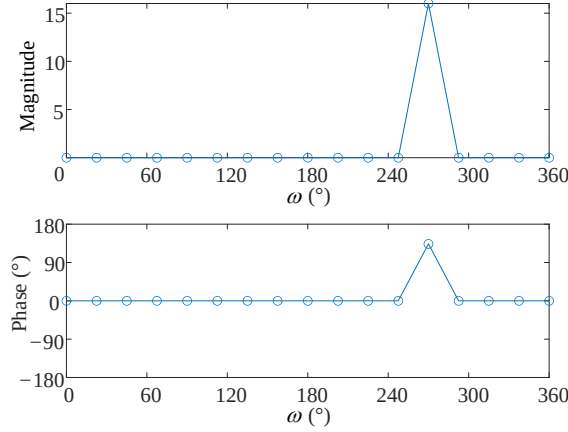


Fig. 4 N -point DFT of an N -point sequence.

Additionally, since there is only one frequency point with a value in the spectrum, it is impossible to analyze the amplitude and phase relationship between the target and its surrounding units. If higher resolution spectra are required, the DTFT sampling rate must be increased, i.e., oversampling. This is equivalent to zero-padding signal in the time domain^[33].

Zero-padding in the time domain means performing an L -point DFT on a sequence of length N , where $L > N$. This requires appending $L - N$ zeros to the end of the sequence. From the DTFT formula, it can be seen that an L -point DTFT and an N -point DTFT are the same because all terms for $n > N$ are zero, i.e.,

$$X(\omega) = \text{DTFT}\{x[n]\} = \sum_{n=0}^{L-1} x[n]e^{-jn\omega} = \sum_{n=0}^{N-1} x[n]e^{-jn\omega} \quad (49)$$

Therefore, zero-padding in the time domain does not add additional frequency information, nor does it alter the original frequency components. However, an L -point DFT is equivalent to performing L -point sampling on the result of the DTFT. Compared to the original N -point sampling, it includes more sampling points and has a higher sampling frequency. Thus, an L -point DFT contains more detailed frequency information, i.e.,

$$\begin{aligned} X[k] &= \text{DFT}\{x[n]\} = X(\omega) \big|_{\omega=2\pi k/L} \\ &= A \exp\{j\phi_{\text{IF}}\} P_L \left(\frac{2\pi}{L} k - \omega_{\text{IF}} \right) \end{aligned} \quad (50)$$

Then, the new sampling interval is written as

$$T_{s,L} = \frac{2\pi}{L} = \frac{N}{L} \cdot \frac{2\pi}{N} \quad (51)$$

The sampling interval is the coefficient before time k . When the sampling interval decreases while k remains an integer variable, the sampling frequency increases, which is referred to as oversampling. Therefore, it can be concluded that time-domain zero-padding and frequency-domain oversampling are equivalent.

Oversampling does not change the radar's resolution capability, nor does it alter the maximum detection range, as no additional information is introduced by oversampling. However, the length of the resolution unit changes. This is because the result of L -point DFT has a length of L , and the range of the frequency axis ω remains $[0, 2\pi)$, but the number of sampling points or the number of frequency intervals becomes L . Therefore, the frequency intervals are more densely packed. The frequency axis corresponds to the distance axis, so the number of resolution units also becomes L . Since the maximum detectable distance remains unchanged, the length of each resolution unit decreases.

For example, performing 2-times oversampling on the result of the DTFT, i.e., $L = 2N$, yields the result shown in Fig. 5(a), while performing 10-times oversampling on the result of the DTFT, i.e., $L = 10N$, yields the result shown in Fig. 5(b).

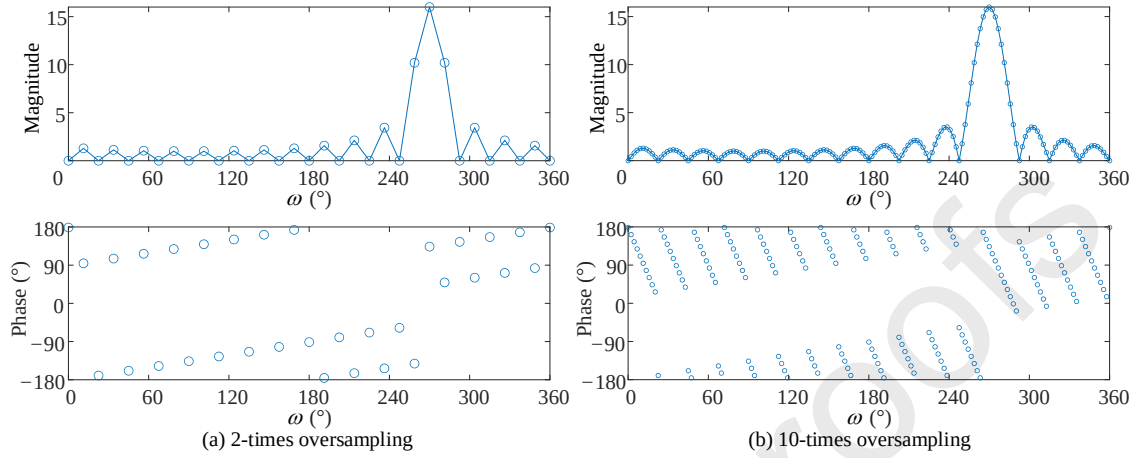


Fig. 5 Oversampling performed on result of DTFT.

The higher the oversampling factor, the finer the results of the DFT, making them closer to the DTFT. In Ref.[33], 5-times and 10-times oversampling were conducted, yielding similar results.

Previously, the condition under which the phase around the target does not undergo abrupt changes was given by Eq. (45), where ω_{IF} is the angular value corresponding to the target. Since the angular axis of the DFT is discrete, this condition changes to

$$\omega_{IF} - \frac{2\pi}{N_s} < \frac{2\pi}{L} k \leq \omega_{IF} + \frac{2\pi}{N_s} \quad (52)$$

Assume the oversampling factor is a , then $L = aN_s$, and the condition becomes

$$\omega_{IF} - \frac{2\pi}{N_s} < \frac{2\pi}{aN_s} k \leq \omega_{IF} + \frac{2\pi}{N_s} \Rightarrow \frac{\omega_{IF} a N_s}{2\pi} - a < k \leq \frac{\omega_{IF} a N_s}{2\pi} + a \quad (53)$$

where k is an integer variable, taking values in the range $[0, L)$.

In the above condition, the restricted range of k has a length of $2a$, centered on the distance of the target, which can be any real number. To ensure that there is at least one frequency component on both sides of the target within two resolution units, that is, to ensure that the phase changes continuously in the resolution units on both sides of the target, it is necessary to ensure that the interval of length $2a$ slides along the real axis, and there are always at least three points in the interval. Therefore, the oversampling factor a needs to be greater than 1.5. Whether it can equal 1.5 depends on the method for approximating the target's position, as the target may not always align precisely with a resolution interval. When the target is located in the middle of two scales, the above inequality cannot be equated if the target is approximated as a smaller scale. If the target is approximated as a larger scale, then the above inequality can be equated.

Under the premise of satisfying the above conditions, the phase variation can be further analyzed. According to the phase variation in Eq. (35), the phase difference between the target and adjacent cell is calculated by

$$\begin{aligned}
\Delta\varphi_p &= \varphi_p\left(\omega + \frac{2\pi}{L}\right) - \varphi_p(\omega) \\
&= -\frac{1}{2} \cdot \frac{2\pi}{L} (N_s - 1) = -\frac{\pi}{a} \left(1 - \frac{1}{N_s}\right)
\end{aligned} \tag{54}$$

3.3. Likelihood function based on phase characteristic correction

Generally, in TBD scenarios, the target can influence the amplitude of several resolution cells in vicinity, and the extent of this influence is related to the blurring factor. For instance, in Refs.[27] and [34], the amplitude characteristics of a target are described using a diffusion function, where the amplitude on the measurement plane diffuses outward from the position corresponding to the target. In the measurement space, the farther a cell is from the target, the lower its amplitude. This aligns with the oversampling results discussed in the previous section. In other words, in the absence of noise, a linearly varying phase with equal differences is valid within a certain range around the target.

The amplitude is essentially the modulus of a complex number. In TBD scenarios, the real and imaginary components of the signal in each cell are influenced by noise, leading to deviations in the signal phase from its true value. In the cell where the target is located, the signal amplitude is relatively high, making it less susceptible to noise. However, in cells farther from the target within the measurement space, where the signal amplitude is inherently low, the phase is more easily affected by noise, making the phase information in these distant cells difficult to utilize. Furthermore, based on the derivation in the previous section, the signal phase varies continuously only within a certain range around the target, and this range is related to the oversampling factor. A higher oversampling factor increases the number of resolution cells influenced by the target, corresponding to a larger blurring factor in the diffusion function. Hence, oversampling and blurring factor are closely linked. Although the blurring factor can vary, the target signal can influence at least one surrounding cell in TBD scenarios. Therefore, there will be always at least one cell around the target where the phase varies continuously.

For a target on the range Doppler resolution unit, assume its coordinates are (r, d) , where r and d represent the range and Doppler of the target, respectively. After rounding, coordinates become $(r_{\text{rod}}, d_{\text{rod}})$. Without considering noise, the target's phase should satisfy

$$\varphi_z(r-1, d) - \varphi_z(r, d) = \varphi_z(r, d) - \varphi_z(r+1, d) \tag{55}$$

where z represent the measurement corresponding to a given coordinate on the measurement plane, which is a complex number, and

$$\varphi_z(r, d) = \varphi[z(r_{\text{rod}}, d_{\text{rod}})] \tag{56}$$

After simplification, we can obtain

$$\varphi_z(r, d) = \frac{1}{2} [\varphi_z(r-1, d) + \varphi_z(r+1, d)] \tag{57}$$

To quantify the degree of deviation between the actual value and model, the PDCM (Phase Deviation from Central Mean) is defined by

$$\varphi_{\text{PDCM}}(r, d) \triangleq \varphi_z(r, d) - \frac{1}{2} [\varphi_z(r-1, d) + \varphi_z(r+1, d)] \tag{58}$$

The phase angle of a single resolution unit can take any value within the range $[-\pi, \pi]$, and the average phase angles of two resolution units can also take any values within the range $[-\pi, \pi]$. The PDCM is their difference, so its value range is $[-2\pi, 2\pi]$. The phase of a complex number is the angle of a vector, and the angle between two vectors should not exceed π . If it exceeds π , the angle should be corrected using modulo operation. Assuming the uncorrected PDCM is φ'_{PDCM} , then the corrected one is denoted by

$$\varphi_{\text{PDCM}} = \text{mod}(\varphi'_{\text{PDCM}} + \pi, 2\pi) - \pi \quad (59)$$

For the units affected by the target, their phase changes linearly. Therefore, PDCM approaches zero or $\pm\pi$, while the phase of the noise is determined by unrelated real and imaginary noise components, distributed uniformly in the range $[-\pi, \pi]$. Although this phase characteristic is specific to the target, noise may also accidentally conform to this characteristic. Therefore, when designing the likelihood function, targets that do not match this characteristic should be suppressed, while those that conform should not be enhanced excessively. Otherwise, noise may be amplified. Hence, the target phase likelihood function is defined as

$$g_{\varphi}(\mathbf{x}_{k|k-1}) \triangleq \exp\left\{-\frac{\varphi_{\text{PDCM}}^2}{\eta^2}\right\} + \exp\left\{-\frac{(\varphi_{\text{PDCM}} - \pi)^2}{\eta^2}\right\} + \exp\left\{-\frac{(\varphi_{\text{PDCM}} + \pi)^2}{\eta^2}\right\} \quad (60)$$

where η represents the standard deviation of PDCM. Thus, the overall likelihood function after phase correction is represented as

$$g_{z,\varphi}(\mathbf{x}_{k|k-1}) = g_{\varphi}(\mathbf{x}_{k|k-1}) \prod_m \frac{\varphi_m(\mathbf{z}_m, \mathbf{x}_{k|k-1})}{\phi_m(\mathbf{z}_m)} \quad (61)$$

After determining the motion model and likelihood function, the LMB-TBD filter is used for data processing obtain the target's state within range-Doppler resolution cells. Finally, the actual range and Doppler values are derived by Eqs. (14) and (15). Thus, according to the descriptions in Sections 3.1, 3.2 and 3.3, the proposed algorithm can be referred to as the RUM-PDCM-LMB-TBD algorithm. The implementation the filter uses a Sequential Monte Carlo (SMC) method, where each target is represented by a series of particles with weights $\{(w^{(\ell,j)}, x^{(\ell,j)})\}$. The pseudocode for the entire filtering process is given in Algorithm 1.

The proposed algorithm is a back-end data processing method whose core principle is to capitalize on the distinctive phase characteristics of targets in the RD domain to enhance the discrimination between targets and clutter. Since the algorithm is built upon the RFS framework, the proposed TBD algorithm possesses strong inherent adaptability. The framework remains applicable even if the phase characteristics of targets in the RD map are more complex. This is because the phase likelihood function within the RFS algorithm can be redesigned to capture more intricate phase or amplitude behaviors, allowing it to suit a wider range of scenarios beyond the simple linear model.

Algorithm 1. Pseudocode for integration of RUM-PDCM-LMB-TBD algorithm.

Input: $\{r_{k-1}^{(\ell)}, p_{k-1}^{(\ell)}\}_{\ell \in \mathbb{L}_{k-1}}$

Output: $\{r_k^{(\ell)}, p_k^{(\ell)}\}_{\ell \in \mathbb{L}_k}$

Step 1. prediction for birth targets

1. for $\ell = L_1 : L_N$
 2. $r_{k,B}^{(\ell)}$ is given by the birth model
 3. for $j = 1 : J^{\ell}$
 4. $p_{k,B}^{(\ell)} = \sum_j \tilde{w}_{k,B}^{(\ell,j)} \delta_{x_{k,B}^{(\ell,j)}}(x)$, $\ell \in \mathbb{B}_k$
 5. where $x_{k,B}^{(\ell,j)} \sim b_k^{(\ell)}(\cdot | z_k)$, $\tilde{w}_{k,B}^{(\ell,j)} = w_{k,B}^{(\ell,j)} / \sum_j w_{k,B}^{(\ell,j)}$ and
-

$$w_{k,B}^{(\ell,j)} = p_{k,B}(x_{k,B}^{(\ell,j)}) / b_k^{(\ell)}(x_{k,B}^{(\ell,j)} | z_k)$$

6. end

7. end

Step 2. prediction for survival targets

8. for $\ell = L_1 : L_N$

9. for $j = 1 : J^\ell$

10. $r_{k|k-1,S}^{(\ell)} = r_{k-1}^{(\ell)} \sum_j w_{k-1}^{(\ell)} p_{s,k}(x^{(\ell,j)}), \ell \in \mathbb{L}_{k-1}$

11. $p_{k|k-1,S}^{(\ell)} = \sum_j \bar{w}_{k|k-1,S}^{(\ell,j)} \delta_{x_{k|k-1}^{(\ell,j)}}(x), \ell \in \mathbb{L}_{k-1}$

12. where $x_{k|k-1}^{(\ell,j)} \sim q_{k|k-1}^{(\ell)}(\cdot | x_{k-1}^{(\ell)}), \bar{w}_{k|k-1,S}^{(\ell,j)} = w_{k|k-1,S}^{(\ell,j)} / \sum_j w_{k|k-1,S}^{(\ell,j)}$

13. $w_{k|k-1,S}^{(\ell,j)} = w_{k|k-1}^{(\ell,j)} f_{k|k-1}(x_{k|k-1}^{(\ell,j)} | x_{k-1}^{(\ell,j)}) p_{s,k}(x^{(\ell,j)}) / q^{(\ell)}(x_{k|k-1,S}^{(\ell,j)} | x_{k-1}^{(\ell,j)})$

14. end

15. end

16. $f_{k|k-1}$ is given by the motion model proposed in Section 3.1.

Step 3. merging of birth and survival predictions

17. $\mathbb{L}_k = \mathbb{L}_{k-1} \cup \mathbb{B}_k$

18. for $\ell = L_1 : L_N$

19. $\{r_{k|k-1}^{(\ell)}\}_{\ell \in \mathbb{L}_k} = \{r_{k,B}^{(\ell)}\}_{\ell \in \mathbb{B}_k} \cup \{r_{k|k-1,S}^{(\ell)}\}_{\ell \in \mathbb{L}_{k-1}}$

20. $\{p_{k|k-1}^{(\ell)}\}_{\ell \in \mathbb{L}_k} = \{p_{k,B}^{(\ell)}\}_{\ell \in \mathbb{B}_k} \cup \{p_{k|k-1,S}^{(\ell)}\}_{\ell \in \mathbb{L}_{k-1}}$

21. end

Step 4. update

22. for $\ell = L_1 : L_N$

23. for $j = 1 : J^\ell$

24. $p_k^{(\ell)} = \sum_j w_{k|k-1}^{(\ell,j)} g_{z,\varphi}(x_{k|k-1}^{(\ell,j)}) \delta_{x_{k|k-1}^{(\ell,j)}}(x) / \rho_k^{(\ell)}, \ell \in \mathbb{L}_k$

25. where $\rho_k^{(\ell)} = \sum_j w_{k|k-1}^{(\ell,j)} g_{z,\varphi}(x_{k|k-1}^{(\ell,j)})$

26. end

27. $r_k^{(\ell)} = r_{k|k-1}^{(\ell)} \rho_k^{(\ell)} / [1 - r_{k|k-1}^{(\ell)} + r_{k|k-1}^{(\ell)} \rho_k^{(\ell)}], \ell \in \mathbb{L}_k$

28. end

29. $g_{z,\varphi}$ is given by the likelihood model in Section 3.3.

Step 5. pruning and merging

30. Prune targets with excessively low existence probabilities.

31. Merge targets that are too close in position.

The computational complexity of the proposed algorithm is primarily concentrated in the prediction and update steps. According to the pseudocode in Algorithm 1, the computational complexity of the prediction step of the proposed RUM-PDCM-LMB-TBD algorithm is $O(NJ)$, where N is the number of targets and J is the maximum number of particles per target. The RUM model in the proposed algorithm does not affect the complexity of the prediction step; hence, the prediction step complexity of the conventional LMB-TBD algorithm is also $O(NJ)$.

As indicated by the pseudocode of the update step in Algorithm 1, the proposed algorithm requires updating the weight of each particle for each target. The update of each particle involves computing the likelihood function given in equation Eq. (61). Assuming M measurements are influenced by the target, the likelihood function for each particle needs to be computed $M+1$ times: M times for the amplitude likelihood and once for the phase likelihood. After updating all particle weights, the existence probability of each target must be also updated. Therefore, the computational complexity of the update step for the proposed algorithm is $O(NJ(M+1)+N)$. In contrast, the conventional LMB-TBD algorithm does not incorporate the phase likelihood function, so its update step complexity is $O(NJM+N)$.

Combining the prediction and update steps, the overall complexity of the proposed RUM-PDCM-LMB-TBD algorithm is $O(NJ(M+2)+N)$, while the overall complexity of the conventional LMB-TBD algorithm is $O(NJ(M+1)+N)$. Thus, the complexity of the proposed algorithm is slightly higher than that of the conventional algorithm. In theory, as the input size grows large, the lower-order terms in the complexity can be neglected. Therefore, the two algorithms are asymptotically equivalent in theoretical complexity, i.e., $O(NJM)$. However, in practical implementations, these minor differences still need to be considered.

4. Observability analysis of bistatic model

The target's range and Doppler data are obtained in Section 3, but they still contain many false alarms and clutter. Additionally, since only a single passive receiving station is used, it can only observe the bistatic range and bistatic Doppler of the target, making it impossible to accurately locate the target's position in Cartesian coordinates. This constitutes an incomplete measurement, requiring target tracking under angle-deficient conditions. First, the observation model under bistatic observations is derived in Section 4.1. Then, Section 4.2 introduces the principles of system observability and degree of observability. Finally, Section 4.3 derives the formula for the system's degree of observability under bistatic observations and analyzes the conditions for observability.

4.1. Bistatic observation model

In this section, the target state vector used is $[x; v_x; y; v_y]^T$, representing the position and velocity of the target in the Cartesian coordinate system. Due to processing point measurements, the observation model is no longer the model used in TBD and needs to be recalculated.

Assuming the transmitting station is T , the target is O , and the receiving station is R , with the target velocity vector as $\mathbf{v}$, the velocity of the target relative to the transmitting station can be understood as the projection of the target's velocity onto the line OT . The length of this projection can be calculated using the inner product, i.e.,

$$v_1 = \frac{\mathbf{OT} \cdot \mathbf{v}}{\|\mathbf{OT}\|} = \frac{(\mathbf{X}_T - \mathbf{X}_O) \cdot \mathbf{v}}{\|\mathbf{X}_T - \mathbf{X}_O\|} \quad (62)$$

$$v_2 = \frac{\mathbf{OR} \cdot \mathbf{v}}{\|\mathbf{OR}\|} = \frac{(\mathbf{X}_R - \mathbf{X}_O) \cdot \mathbf{v}}{\|\mathbf{X}_R - \mathbf{X}_O\|} \quad (63)$$

where $\mathbf{X}_T$, $\mathbf{X}_R$ and $\mathbf{X}_O$ represent the coordinates of the transmitting station, receiving station and target, respectively. Substituting these into Eq. (9), the total frequency shift from the transmitting station to the receiving station can be obtained by

$$f_d = \frac{1}{\lambda_0} \left(\frac{(\mathbf{X}_T - \mathbf{X}_O) \cdot \mathbf{v}}{\|\mathbf{X}_T - \mathbf{X}_O\|} + \frac{(\mathbf{X}_R - \mathbf{X}_O) \cdot \mathbf{v}}{\|\mathbf{X}_R - \mathbf{X}_O\|} \right) \quad (64)$$

where the carrier wavelength λ_0 is a constant, which can be moved to the left side, namely,

$$v_r = f_d \cdot \lambda_0 = \frac{(\mathbf{X}_T - \mathbf{X}_O) \cdot \mathbf{v}}{\|\mathbf{X}_T - \mathbf{X}_O\|} + \frac{(\mathbf{X}_R - \mathbf{X}_O) \cdot \mathbf{v}}{\|\mathbf{X}_R - \mathbf{X}_O\|} \quad (65)$$

where v_r represents the bistatic velocity, which is the sum of the radial velocity of the target relative to the transmitting station and the radial velocity of the target relative to the receiving station. Next, given that the distance between the transmitter and the receiver is known, the time delay from the transmitting station to the receiving station, i.e., the signal propagation time, is calculated by

$$\tau_{bi} = \frac{\|\mathbf{OT}\| + \|\mathbf{OR}\|}{c} = \frac{\|\mathbf{X}_T - \mathbf{X}_O\| + \|\mathbf{X}_R - \mathbf{X}_O\|}{c} \quad (66)$$

where the speed of light c is a constant, which can be moved to the left side. Thus, we can obtain

$$r_{bi} = \tau_{bi} \cdot c = \|\mathbf{X}_T - \mathbf{X}_O\| + \|\mathbf{X}_R - \mathbf{X}_O\| \quad (67)$$

Finally, the observation equation of the bistatic observation model is obtained, which corresponds to the observation function in Eq. (5), i.e.,

$$h(\mathbf{x}) = \begin{bmatrix} r_{bi} \\ v_r \end{bmatrix} = \begin{bmatrix} \|\mathbf{X}_T - \mathbf{X}_O\| + \|\mathbf{X}_R - \mathbf{X}_O\| \\ \frac{(\mathbf{X}_T - \mathbf{X}_O) \cdot \mathbf{v}}{\|\mathbf{X}_T - \mathbf{X}_O\|} + \frac{(\mathbf{X}_R - \mathbf{X}_O) \cdot \mathbf{v}}{\|\mathbf{X}_R - \mathbf{X}_O\|} \end{bmatrix} \quad (68)$$

Assuming the receiving station is located at the origin, i.e., $\mathbf{X}_R = [0, 0]$, Eq. (68) can be further written as the form of Cartesian coordinates, namely,

$$h(\mathbf{x}) = \begin{bmatrix} r_{bi} \\ v_r \end{bmatrix} = \begin{bmatrix} \sqrt{(x_t - x)^2 + (y_t - y)^2} + \sqrt{x^2 + y^2} \\ \frac{(x_t - x)v_x + (y_t - y)v_y}{\sqrt{(x_t - x)^2 + (y_t - y)^2}} + \frac{-v_x x - v_y y}{\sqrt{x^2 + y^2}} \end{bmatrix} \quad (69)$$

It should be here noted that since the observation equation Eq. (69) cannot be directly expressed in the form of matrix multiplication as in Eq. (7), the system is a nonlinear system.

4.2. Principles of system observability and observable degree

In the previous section, the nonlinear observation model under the bistatic scenario was presented, and the observation matrix for the first-order approximation was derived. It is known that passive radar can measure the target's bistatic time delay and Doppler shift, from which the target's bistatic range and bistatic velocity can be obtained. However, when using the passive measurement information for tracking and filtering, some uncertainties still remain: under what conditions can the target's complete state be determined using this passive measurement information, and to what extent can the target's complete state be determined? These questions can all be classified as the target observability problem, and studying conditions for target observability is key to addressing such issues.

Ignoring noise, a discrete nonlinear system can be represented by

$$\begin{cases} \mathbf{x}_k = \mathbf{f}_{k|k-1}(\mathbf{x}_{k-1}) \\ \mathbf{z}_k = \mathbf{h}_k(\mathbf{x}_k) \end{cases} \quad (70)$$

A commonly used method for approximating a nonlinear function is Jacobian matrix^[35], which is an analytical approximation method. However, if the nonlinear function is discontinuous, this method cannot be applied. By replacing the nonlinear function with its corresponding Jacobian matrix, the system can be approximated as

$$\begin{cases} \mathbf{x}_k = \mathbf{F}_{k|k-1} \mathbf{x}_{k-1} \\ \mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k \end{cases} \quad (71)$$

If the measurement is a one-dimensional scalar, the state vector is n -dimensional, and n groups of measurement data $z_1, z_2, \dots, z_n$ are obtained, all measurement equations can be written as

$$\begin{cases} z_1 = \mathbf{H}_1 \mathbf{x}_1 \\ z_2 = \mathbf{H}_2 \mathbf{F}_{2|1} \mathbf{x}_1 \\ \vdots \\ z_n = \mathbf{H}_n \mathbf{F}_{n|n-1} \mathbf{F}_{n-1|n-2} \cdots \mathbf{F}_{2|1} \mathbf{x}_1 \end{cases} \quad (72)$$

In matrix form, Eq. (72) can be rewritten as

$$\begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} = \begin{bmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \mathbf{F}_{2|1} \\ \vdots \\ \mathbf{H}_n \mathbf{F}_{n|n-1} \mathbf{F}_{n-1|n-2} \cdots \mathbf{F}_{2|1} \end{bmatrix} \mathbf{x}_1 = \mathbf{F} \mathbf{x}_1 \quad (73)$$

By Eq. (73), it can be seen that if the matrix $\mathbf{F}$ is invertible or full rank, the target's state can be uniquely determined from measurements, and the system is said to be observable. If measurements are multidimensional vectors, the required number of measurements decreases accordingly, and this criterion remains valid.

However, whether the matrix satisfies full rank can only indicate whether the system is observable, and cannot distinguish the strength of observability under noise when estimating the target state. Therefore, a measure is needed to describe the strength of the system observability. In an observable system, let $\mathbf{h}(\mathbf{x}_k) = [h_1^T(\mathbf{x}_1), h_2^T(\mathbf{x}_2), \dots, h_k^T(\mathbf{x}_k)]^T$, $\mathbf{Z} = [z_1^T, z_2^T, \dots, z_k^T]^T$. Using the nonlinear equation group $\mathbf{h}(\mathbf{x}_k) = \mathbf{Z}$ and solving it via Newton iterative method, the following approximate solution can be obtained by

$$\mathbf{x}_k = \mathbf{x}_k^* + \mathbf{F}^{-1}(k, 1)[\mathbf{Z} - \mathbf{h}(\mathbf{x}_k^*)] \quad (74)$$

where $\mathbf{x}_k^*$ is the initial state of $\mathbf{x}_k$. Therefore, the strength of the system observability is reflected in whether $\mathbf{F}(k,1)$ is close to singular or ill-conditioned. In fact, for linear systems, eigenvalues and eigenvectors of Gramer matrix $\mathbf{F}^T \mathbf{F}$ fully describe the system observability. The same applies to nonlinear systems. However, it is generally not easy to analytically obtain eigenvalues and eigenvectors. If $\mathbf{x}_k^*$ is the ground truth, from Eq. (74), it can be seen that matrix $\mathbf{F}^{-1}(k,1)$ reflects the sensitivity of the deviation $\mathbf{x}_k - \mathbf{x}_k^*$ to the influence of measurement noise $\mathbf{Z} - h(\mathbf{x}_k^*)$. If the covariance of the measurement noise is $\mathbf{R}$, then the covariance of the state is written as

$$\text{cov}(\mathbf{X}_k - \mathbf{X}_k^*) = (\mathbf{F}^T \mathbf{R}^{-1} \mathbf{F})^{-1} \quad (75)$$

where $\mathbf{F}^T \mathbf{R}^{-1} \mathbf{F}$ is called the Fisher information matrix. Since $\det(\mathbf{F}^T \mathbf{R}^{-1} \mathbf{F})^{-1}$ is a measure of the volume of the hyperellipsoid of the state covariance, $|\det \mathbf{F}|$ can be also considered to have the property of accuracy measurement. Thus, the observable degree of nonlinear systems can be defined as $\rho = |\det \mathbf{F}|$, which is used to compare the relative strength of observability. When $\rho = 0$, the matrix is not invertible, indicating that the target state is unobservable. The range of the observable degree ρ is $[0,1]$. For example, if the state is $\mathbf{x} = [x, v_x, y, v_y]^T$ and the measurement includes all components of the target's state, the observable degree is $\rho = |\det \mathbf{F}| = |\det \mathbf{H}| = 1$, where $\mathbf{H}$ is the unit matrix. Similarly, if the measurement only includes coordinates x and y , the observable degree is also calculated as one.

4.3. Observable degree for moving targets under bistatic observation

In a two-dimensional plane, the position of the target requires at least two parameters for representation, and the magnitude and direction of velocity also requires at least two parameters. Suppose the state vector of the target is $\mathbf{x} = [x, v_x, y, v_y]^T$. It can completely represent the target's position, velocity and direction information. Since the measurement is a two-dimensional vector, determining the four-dimensional state requires at least two moments of measurements. When calculating observable degree, the state transition equation is generally a linear equation, and $\mathbf{F}$ is defined as

$$\mathbf{F} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (76)$$

The observable matrix is expressed as

$$\mathbf{F} = \begin{bmatrix} \mathbf{H}_k \\ \mathbf{H}_{k+1} \mathbf{F} \end{bmatrix} \quad (77)$$

When establishing a coordinate system, set the receiving station as the origin. The line connecting the receiving station and transmitting station is taken as the x axis, and the direction from the receiving station to transmitting station is positive along the x axis. Thus, the coordinates of the receiving and transmitting stations are $(0,0)$ and $(x_t, 0)$, respectively. In this scenario, Eq. (69) provides the corresponding observation equation whose Jacobi matrix is calculated as

$$\mathbf{H} = \frac{\partial h(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial v_x} & \frac{\partial r}{\partial y} & \frac{\partial r}{\partial v_y} \\ \frac{\partial v_r}{\partial x} & \frac{\partial v_r}{\partial v_x} & \frac{\partial v_r}{\partial y} & \frac{\partial v_r}{\partial v_y} \end{bmatrix} \quad (78)$$

where the calculation process of each element of matrix $\mathbf{H}$ is shown in Appendix A.

It is important to note that Eq. (69) omits the time index of the state vector, while $\mathbf{H}_k$ and $\mathbf{H}_{k+1}$ represent the state vector indices at times k and $k+1$, respectively. These two indices are related but represent different variables at two different times. Subsequently, it is only necessary to compute the determinant of $\mathbf{F}$, i.e.,

$$\det(\mathbf{F}) = a_x f_1 + a_y f_2 \quad (79)$$

where the calculation process of variables a_x , a_y , f_1 and f_2 is shown in Appendix B.

Note that $\det(\mathbf{F})$ is obtained by summing two parts. The coefficients of two parts are a_x and a_y , respectively. When the target is located on the line connecting the transmitting and receiving stations at time k , i.e., $y_k = 0$, and $0 < x_k < x_t$, two coefficients are both 0, making the observable degree zero. In this case, the target is unobservable. Additionally, if the target is stationary, in the special case where $v_x = v_y = 0$ at time k , the observation function's second term becomes zero. This is a constant function, and a constant term's partial derivative with respect to any variable is zero. Therefore, the second row of the observable matrix $\mathbf{H}_k$ is zero. Since one row is entirely zero, the observable degree matrix $\mathbf{F}$ is not full rank and the target is not observable. In summary, the conditions for the target to be unobservable are: the target lies on the line segment connecting the transmitting and receiving stations, or the target is stationary. If either of these conditions is met, the target state is unobservable.

5. Numerical experiments

This section will present simulation results to verify the proposed algorithm, including a comparison of the effects before and after improvements to the TBD algorithm, the observability distribution of the bistatic observation model for moving targets, and an analysis of the impact of observability on estimation accuracy, where the evaluation criterion for estimation accuracy is the Optimal Sub-Pattern Assignment (OSPA) metric.

The following experiments are conducted under the scenario where both the transmitter and the receiver are assumed to be fixed. Furthermore, it is assumed that targets do not migrate across range-Doppler resolution cells within a single step. This setup allows us to focus the analysis on the core algorithmic performance without the confounding factors of platform maneuvering or migration effects, thereby creating a controlled setting to directly attribute the tracking performance to the efficacy of the TBD algorithm itself.

5.1. Performance of TBD algorithm

First, in the simulation scenario, there are four targets moving at a constant velocity along straight lines in a Cartesian coordinate system. The initial states, birth times and death times of the targets are listed in Table 1. The targets perform nonlinear motion in the RD (Range-Doppler) space. The ground truth in the RD space is shown in Fig. 6, which is also used as the reference for calculating the OSPA error, where, where “○” and “△” represent the beginning and ending of each track, respectively.

Table 1 Target motion parameters.

Target	Initial state	Birth time	Death time
1	[50; 8; 1500; -50]	1	20
2	[70; 36; 1500; -36]	1	20
3	[50; 22; 1200; -43]	1	20

4 [900; -36; 1300; -27] 1 20

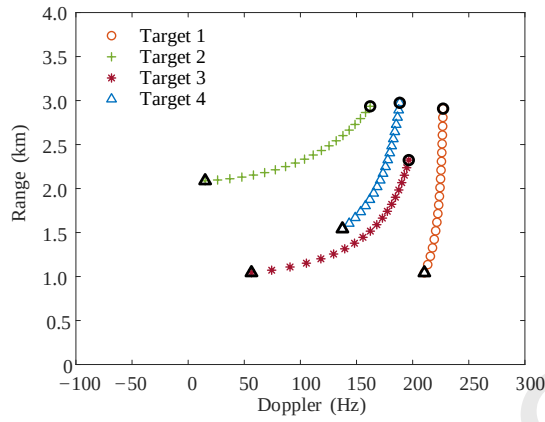


Fig. 6 Ground truth in RD space.

To compare the performance of the algorithm before and after improvement under SNR conditions of 10 dB, 8 dB and 6 dB, 100 Monte Carlo (MC) experiments are conducted. Since the target states are measured in resolution cells, the cutoff distance for OSPA is set to 5, meaning that estimates deviating from more than 5 resolution cells will contribute to cardinality errors rather than location errors. The original LMB-TBD algorithm without improvements is used as the comparison algorithm. The experimental results are shown in Fig. 7, Fig. 8 and Fig. 9, including the mean cardinality estimate, OSPA distance, OSPA location error, and OSPA cardinality error.

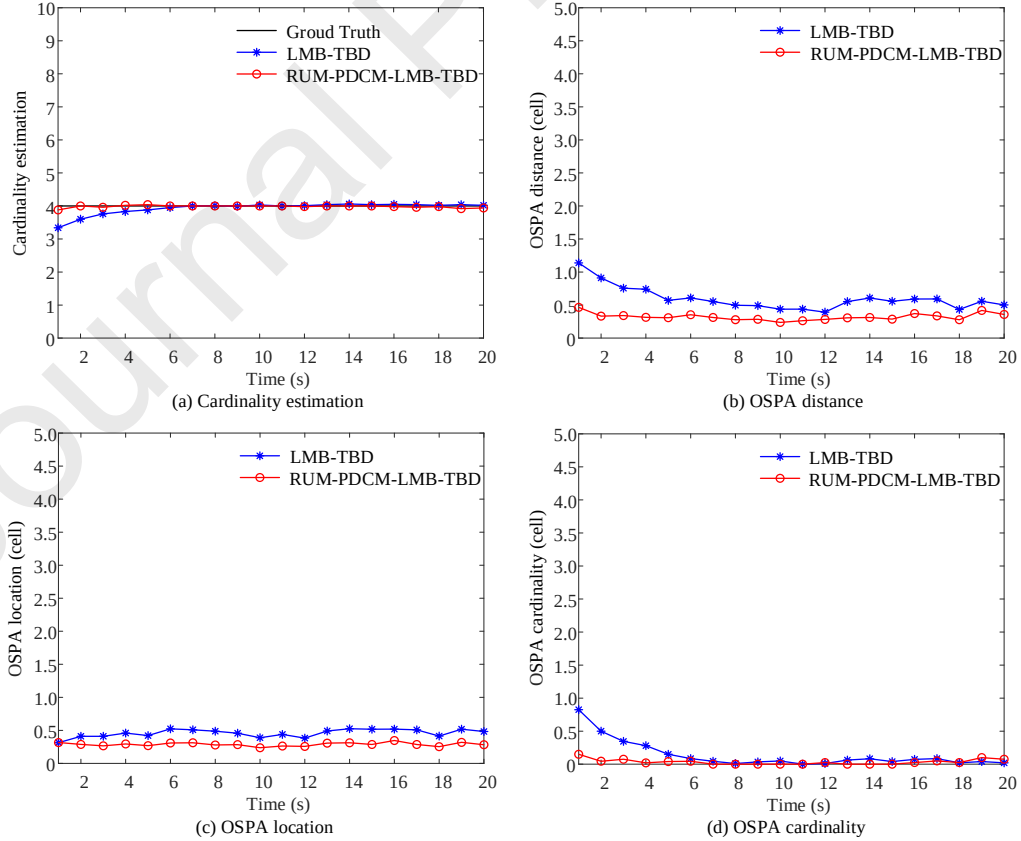


Fig. 7 Experimental results of two comparative algorithms under SNR=10 dB.

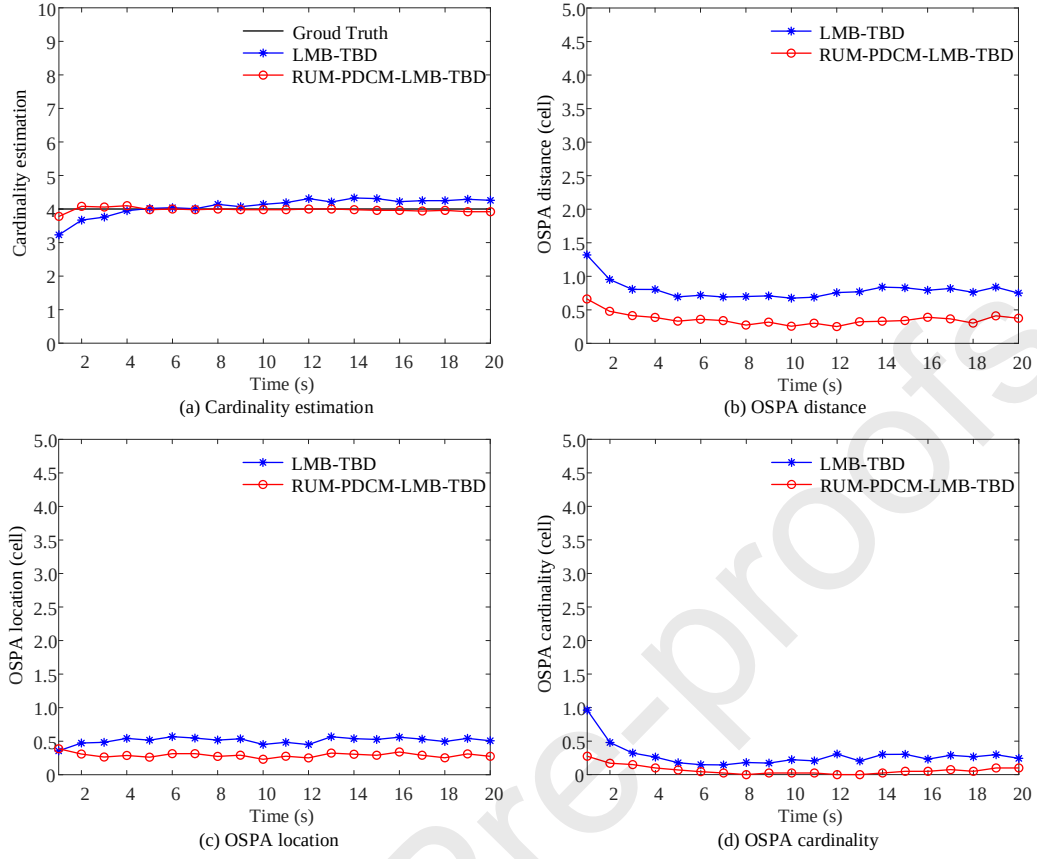


Fig. 8 Experimental results of two comparative algorithms under SNR=8 dB.

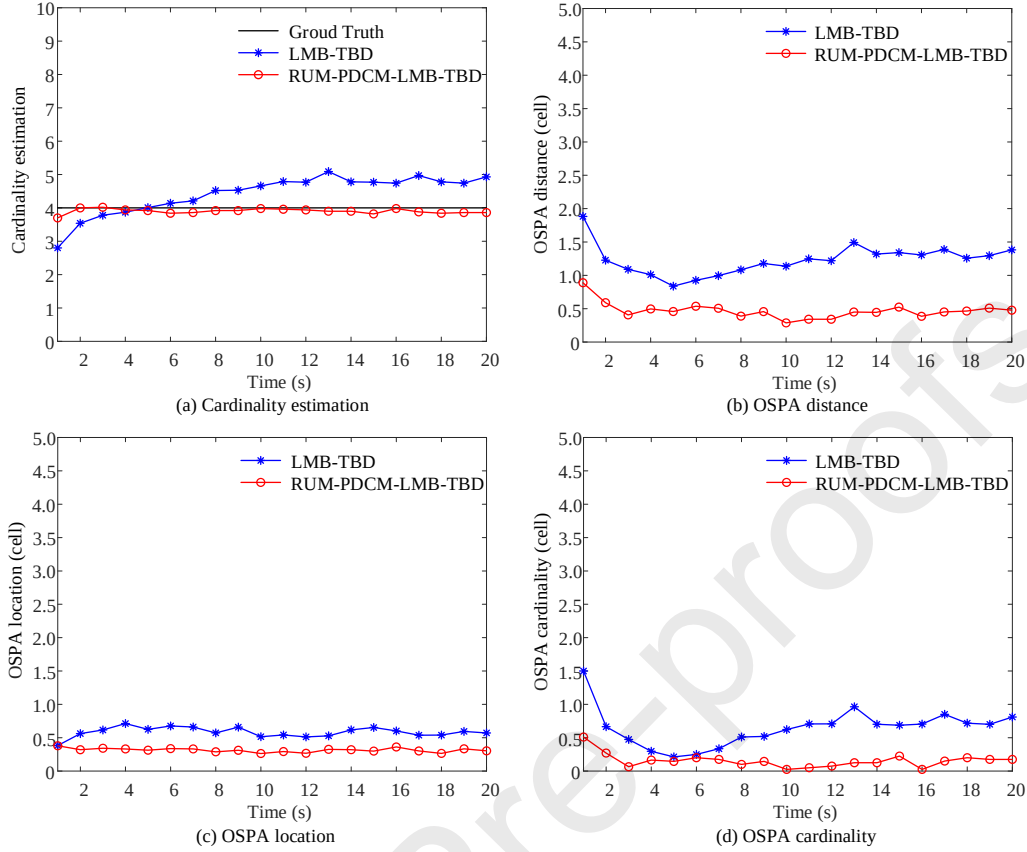


Fig. 9 Experimental results of two comparative algorithms under SNR=6 dB.

Firstly, under the 10 dB condition, as seen in Figs. 7(c) and (d), both algorithms perform well in cardinality estimation and location estimation. However, in Fig. 7(a), the LMB-TBD algorithm slightly underestimates the number of targets during the initial moments, and the overall OSPA is slightly higher than that of the proposed RUM-PDCM-LMB-TBD algorithm. Secondly, under the 8 dB condition, as shown in Fig. 8(a), the LMB-TBD algorithm slightly overestimates the number of targets in the final moments. In contrast, the proposed algorithm, utilizing phase characteristics to suppress clutter, maintains stability in cardinality estimation. Regarding position error in Fig. 8(c), the proposed algorithm significantly outperforms the LMB-TBD algorithm. Overall, as seen in Fig. 8(b), the gap in OSPA between the two algorithms is larger than that under the 10 dB condition. Finally, under the 6 dB condition, as shown in Fig. 9(a), the cardinality estimation of the LMB-TBD algorithm is significantly higher than the true value. For the proposed algorithm, when utilizing phase characteristics, the low signal amplitude under the 6 dB condition makes its phase more susceptible to clutter and noise, leading to a slight underestimation of the target number. However, in terms of position error, Fig. 9(c) shows that the proposed algorithm still outperforms the LMB-TBD algorithm. Overall, as seen in Fig. 9(b), the proposed algorithm is significantly better than the LMB-TBD algorithm.

In conclusion, when the SNR is high, the proposed RUM-PDCM-LMB-TBD algorithm performs comparably to the LMB-TBD algorithm in cardinality estimation but surpasses it in position estimation. When the SNR is low, the proposed RUM-PDCM-LMB-TBD algorithm outperforms the LMB-TBD algorithm significantly in both cardinality and position estimation. However, it is important to note that this performance advantage diminishes in extremely low SNR condition. Under such condition, noise severely corrupts the target's phase characteristics in the RD map, the core phase refinement module of our algorithm struggles to capture a stable and distinctive target signature. Therefore, the algorithm may inadvertently suppress potential targets. Thus, after a large number of simulation experiments, it is advisable to apply the proposed algorithm in environments with a SNR above 3 dB to ensure superior detection performance over traditional method. Therefore, the SNR boundary of the applicability of the proposed algorithm is finally set to 3 dB.

To further evaluate the target detection performance of the proposed method, a Receiver Operating Characteristic (ROC) curve analysis is conducted under 6 dB condition. In this experiment, the proposed algorithm and comparison

algorithm use identical parameter configurations, with only the final detection threshold varied systematically over a wide range. For each threshold, the detection rate and false alarm rate are calculated, collectively forming the ROC curves. As shown in Fig. 10, the proposed RUM-PDCM-LMB-TBD algorithm achieves a higher ROC curve compared to the LMB-TBD algorithm, indicating superior detection performance across different threshold settings. Overall, the proposed algorithm exhibits robust and comprehensive superiority in both state estimation and target detection under challenging low-SNR conditions.

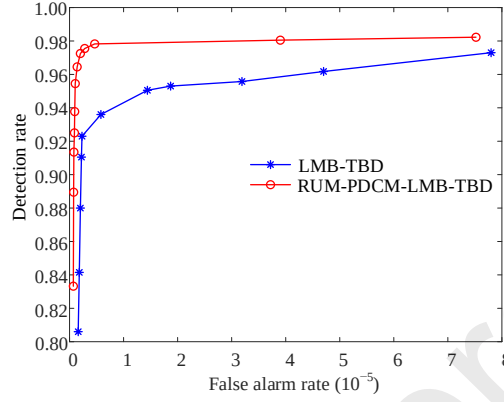


Fig. 10 ROC curve of the algorithm.

In 100 Monte Carlo simulations, the overall runtime of both algorithms, as well as a comparison of their per-step execution time, are provided in Table 2 to analyze the computational efficiency of the proposed algorithm. Both algorithms were tested under identical experimental parameters, with the maximum number of particles per target set to 1200. As analyzed in Section 3.3, the proposed algorithm has slightly higher computational complexity than the conventional LMB-TBD algorithm under the same input scale, which is reflected in the actual runtimes.

Table 2 Average algorithm running time.

Algorithm	Overall runtime (s)	Single step running time (s)
LMB-TBD	1193.57	0.597
RUM-PDCM-LMB-TBD	1343.62	0.672

The computational complexity is mainly determined by the number of targets, measurements and particles. Among these, the number of particles is a configurable parameter. In practice, the proposed algorithm employs a 4-dimensional state vector $\mathbf{x} = [x_r, x_d, v_{x_d}, I]^T$, where x_r , x_d and v_{x_d} represent dimensions related to motion, and I denotes the amplitude dimension, thereby reducing the state dimensionality by one compared to the conventional LMB-TBD algorithm. This reduction enables more efficient exploration of the state space. As a result, fewer particles are required to accurately describe the target state. For example, while the conventional LMB-TBD algorithm requires at least 1200 particles per target, the proposed RUM-PDCM-LMB-TBD algorithm only needs around 1000 particles. Such parameter optimization can improve operational efficiency without compromising tracking performance.

5.2. Observable degree distribution of moving targets

Generally, the stronger the observable degree in a region, the more information the filter can extract from measurements, and the more accurate the target state estimation will be. According to Eqs. (77) and (78), the observable degree ($\rho = |\det \mathbf{F}|$) is related to the determinant of the observable matrix, and $\mathbf{F}$ is composed of $\mathbf{H}$. Therefore, ρ is also related to the state of the target. To understand the observable degree strength in different regions of the plane, it is necessary to separately calculate the observable degree corresponding to different target states.

In the previous derivation of the observable degree expression, the receiving station was set as the origin, and the direction from the receiving station to transmitting station was taken as the positive x -axis, where the transmitting station was positioned at $(x_t, 0)$, assume $x_t = 200$ m. To intuitively visualize how the observable degree varies with the target's full state (position and velocity), a comprehensive analysis is conducted. While the target position can be traversed continuously across the 2D plane, the velocity vector has an infinite number of possible directions. To address this in a representative and fair manner, we selected eight distinct motion directions that are most illustrative, i.e., right: (1 m/s, 0 m/s), left: (-1 m/s, 0 m/s), up: (0 m/s, 1 m/s), down: (0 m/s, -1 m/s), upper-right: ($1/\sqrt{2}$ m/s, $1/\sqrt{2}$ m/s), upper-left: ($-1/\sqrt{2}$ m/s, $1/\sqrt{2}$ m/s), lower-right: ($1/\sqrt{2}$ m/s, $-1/\sqrt{2}$ m/s), and lower-left: ($-1/\sqrt{2}$ m/s, $-1/\sqrt{2}$ m/s). The magnitude of all velocity vectors is unified and set to one to isolate the effect of movement direction.

Within a certain range on the plane, the observable degree for each point is calculated for targets moving in these 8 different directions. The comparison is shown in Fig. 11, which consists of eight subplots. The layout of these subplots intentionally corresponds to the direction of the velocity vector they represent (e.g., the subplot for the “upper-right” velocity is placed in the upper-right corner), with an arrow indicating the speed direction placed in the center of the figure.

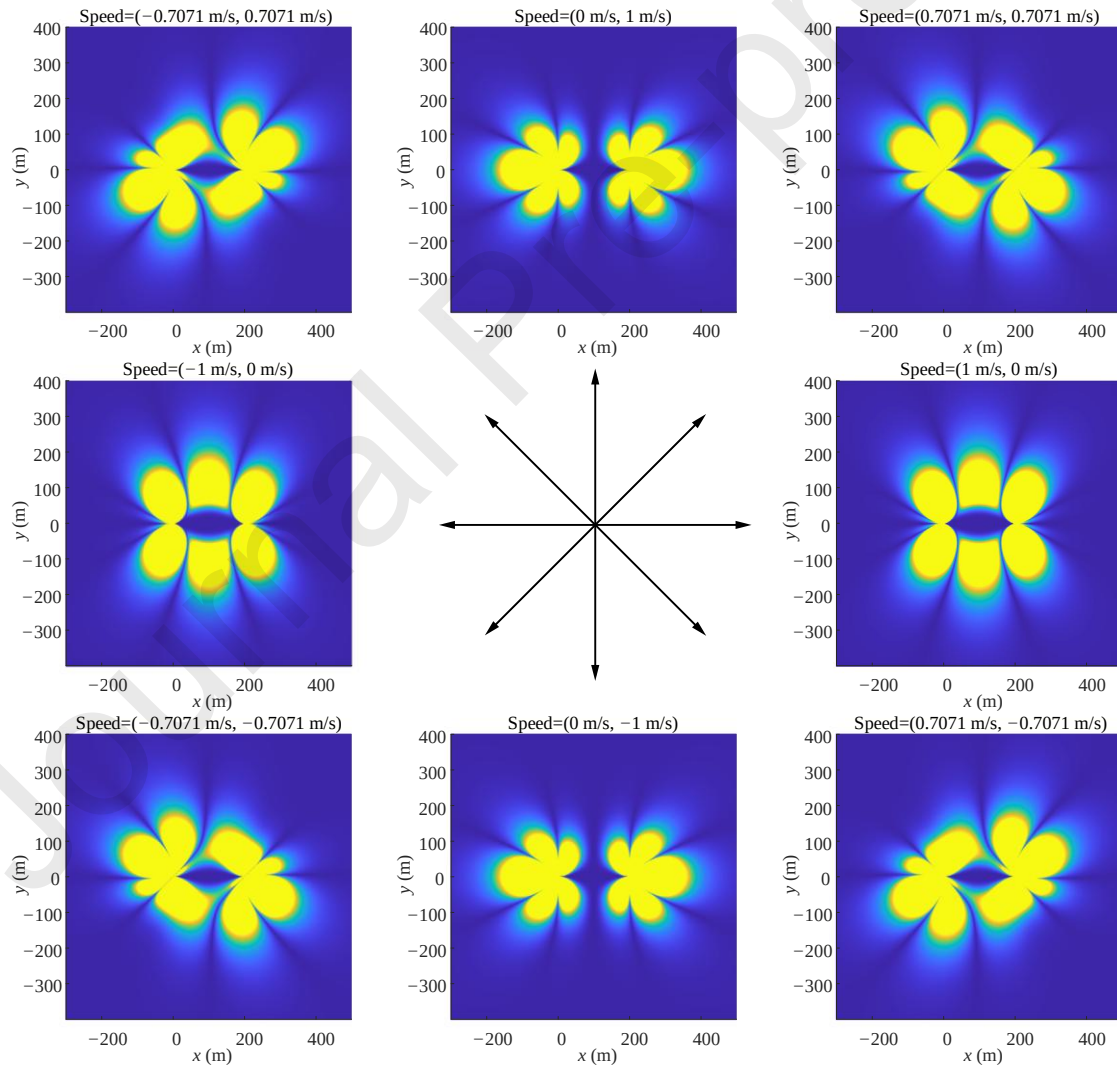


Fig. 11 Comparison of observable degree in different motion directions.

From Fig. 11, it can be observed that, at a uniform velocity magnitude of one, the high observable degree regions are primarily located around two stations, and the velocity direction significantly affects the spatial distribution of the observable degree. Furthermore, the influence of the velocity magnitude is analyzed by selecting a single and representative motion direction (the positive x-axis) and calculating the observable degree across the plane for different speeds. This comparative analysis, shown in Fig. 12, demonstrates that the observable degree increases with the target's velocity.

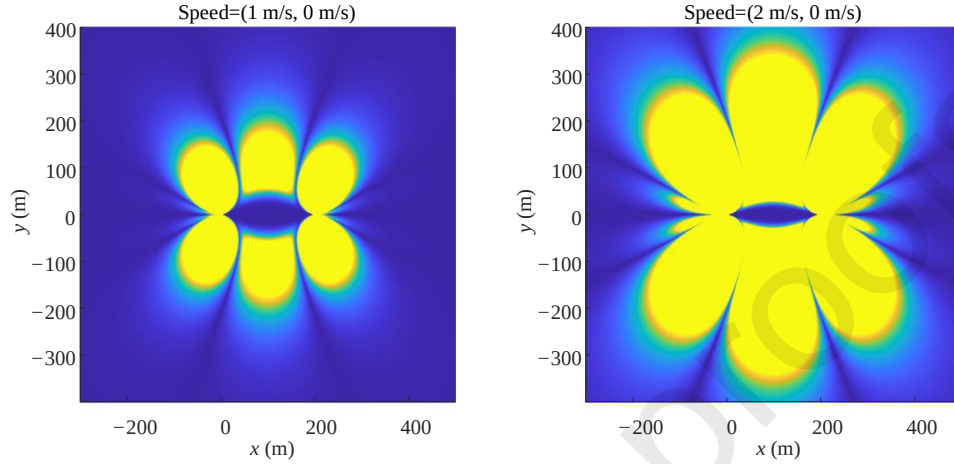


Fig. 12 Comparison of observable degree at different speeds.

5.3. Impact of observable degree on estimation accuracy

In the previous section, the calculation of observable degree utilized Jacobian matrix of the observation function. Similarly, the extended Kalman filter (EKF) uses an analogous approximation method. However, this approximation introduces errors in highly nonlinear scenarios^[36]. Therefore, this work adopts an implementation based on the generalized labeled multi-Bernoulli (GLMB) particle filter to mitigate errors caused by nonlinear factors.

In experiment 1, assume the target starts at position $(-250 \text{ m}, 350 \text{ m})$ and moves in a straight line with an initial velocity of $(10 \text{ m/s}, -10 \text{ m/s})$, with small process noise. The actual trajectory will exhibit some curvature. The initial covariance of the target is set to $\text{diag}(10^2, 1^2, 10^2, 1^2)$. The target's initial state is randomly distributed according to the covariance, and it moves for 50 s. According to the previous observable degree distribution situation, the target will initially move from a bright (high observable degree) region into a dark (low observable degree) region, then, after crossing the line between two stations, it will move from the dark region back into the bright region, and finally exit into the dark region again. The simulated trajectory and estimation error results are shown in Fig. 13.

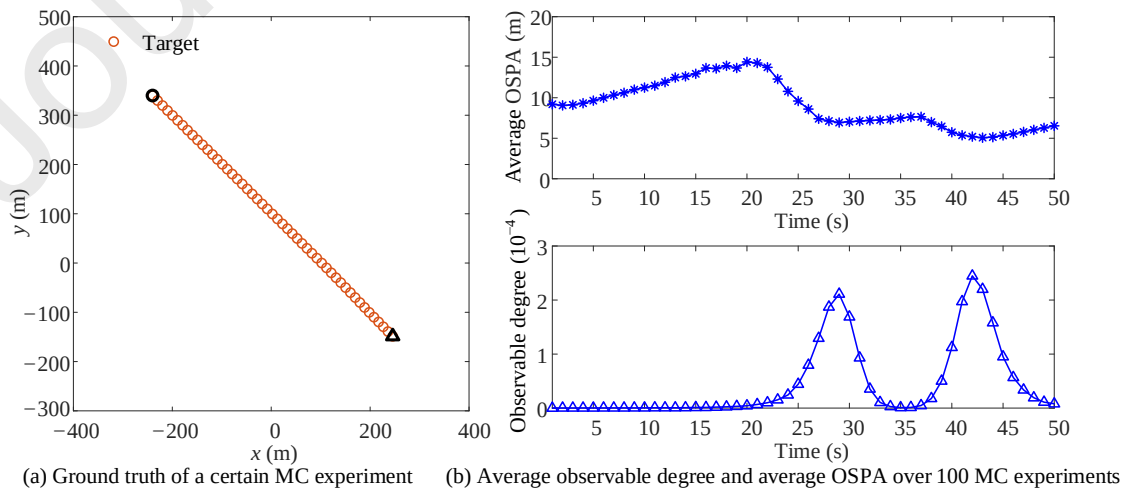


Fig. 13 Experiment 1 on observable degree and estimation accuracy.

The OSPA metric reflects the instantaneous error values. From Fig. 13(b), it can be observed that the error initially increases gradually, followed by two significant decreases. The increase in error during the first 20 s is due to the target being in a region of low observable degree, i.e. the dark region. From 20 s to 30 s, the observable degree steadily increases, resulting in a reduction in error. Around 35 s, the target crosses the unobservable line, reducing observable degree to zero, and the error remains unchanged. From 38 s to 43 s, the target enters the bright region below the axis, causing the estimation error to decrease again. In the final few time steps, the observable degree gradually declines, leading to a slight increase in estimation error. These results indicate a strong correlation between estimation error and observable degree.

In experiment 2, the target's initial position is set to $(-200 \text{ m}, 65 \text{ m})$, with a velocity of $(12 \text{ m/s}, 0 \text{ m/s})$ and a smaller process noise. The target moves for 50 s, and the results are shown in Fig. 14. It can be observed that the target sequentially passes through two regions of high observable degree, where the estimation error significantly decreases at the corresponding positions. Subsequently, as the observable degree gradually decreases, the estimation error increases accordingly.

In experiment 3, the target's initial position is set to $(-200 \text{ m}, 120 \text{ m})$ with an initial velocity of $(12 \text{ m/s}, 0 \text{ m/s})$, and the results are shown in Fig. 15. Throughout the experiment, the observable degree remains consistently low. During the first 17 s, the observable degree slightly increases, resulting in a gradual decrease in estimation error. From 17 s to 35 s, the observable degree fluctuates at a relatively low level, causing the error to remain stable without significant changes. However, from 35 s to 50 s, the observable degree continuously decreases, leading to a faster increase in estimation error. From the above analysis, it can be concluded that when the level of observable degree is too low, the filter struggles to extract sufficient information from measurements, making it challenging to achieve more accurate state estimates.

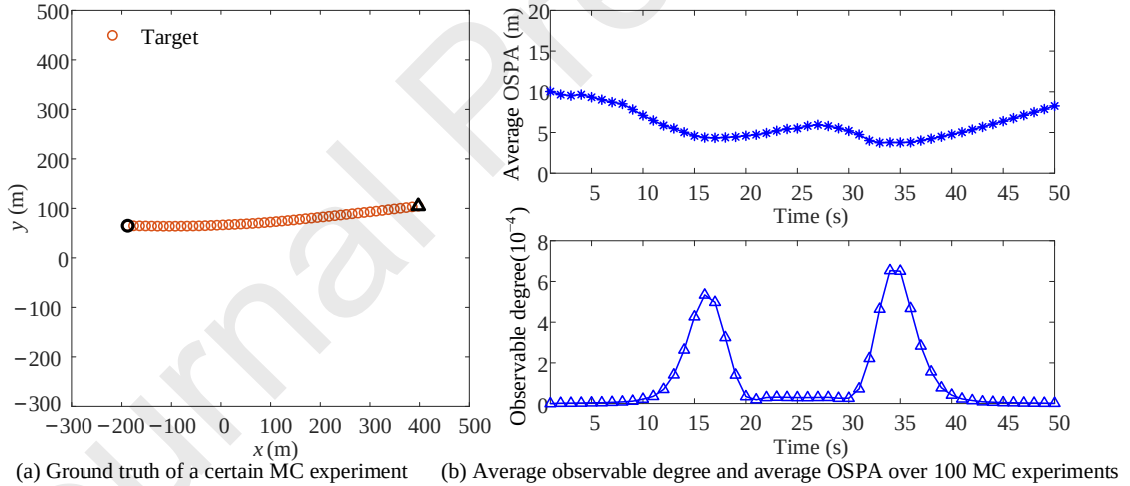


Fig. 14 Experiment 2 on observable degree and estimation accuracy.

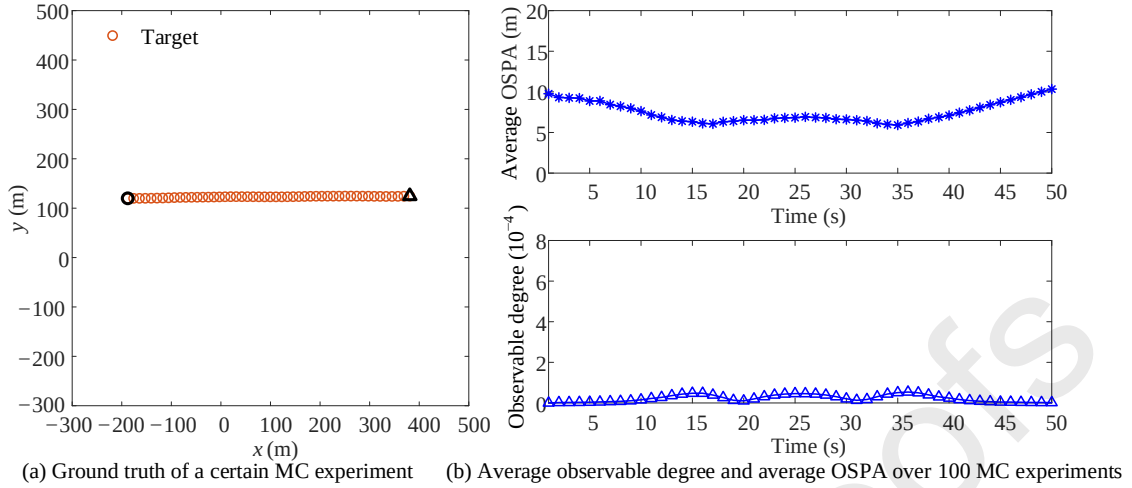


Fig. 15 Experiment 3 on observable degree and estimation accuracy.

Overall, these experiments demonstrate that observable degree can effectively reflect the estimation accuracy of the target state. As long as observable degree remains at a high level, the estimation error will consistently decrease. However, due to the influence of random noise, when observable degree is too low, the estimation error may remain unchanged or even increase.

It is here emphasized that OSPA is measured in the Cartesian coordinate system and represents the positional error under the condition where cardinality errors are not considered. The initial position of the target is random, which results in an initial error. Specifically, since the target's initial position at time 0 is randomly distributed with a standard deviation σ in both x and y directions, the positional error at time 1 s follows a Rayleigh distribution with a parameter of σ and its expected value is $\sigma\sqrt{\pi/2}$. Considering the velocity error and process noise, this value will actually be higher. In other words, without the filtering process, the initial error reflected in the OSPA metric would be at least $\sigma\sqrt{\pi/2}$. In the scenarios of the three experiments above, $\sigma=10$ m, so the initial error is approximately $\sigma\sqrt{\pi/2} \approx 12.5$ m. In experiments 2 and 3, the error over all 50 s remains mostly below the initial error. However, in experiment 1, due to the target remaining in a low observable degree region for an extended period at the beginning, the OSPA metric briefly exceeds the initial error.

In conclusion, as long as the target does not stay in a low observable degree region for an extended period, the OSPA metric will remain at a relatively low value. From the distribution of observable degree discussed in the previous section, it is evident that as long as the target stays within a certain range of the two stations, the estimation error will remain low. Furthermore, the greater the target's velocity, the larger this range becomes.

6. Conclusions

This work focuses on the problem of TBD in passive bistatic observation and proposes a TBD algorithm based on labeled multi-Bernoulli RFS, referred to as the RUM-PDCM-LMB-TBD algorithm. By improving the motion model of the traditional TBD algorithm, the proposed algorithm can effectively utilize Doppler information while avoiding the issue of mutual influence of filtering parameters. Additionally, the likelihood function model is refined by deriving the phase distribution characteristics of the target, which fully exploits the phase information in measurements. This improvement addresses the issues of inaccurate cardinality estimation and state estimation in the original algorithm. Simulation results under three different SNR scenarios demonstrate that the proposed algorithm achieves significantly better estimation performance compared to the original algorithm. Finally, the results of TBD are applied to point target tracking, yielding the final target state. In this process, by analyzing the observability and observable degree of the target, the conditions for observability are provided, and the relationship between observable degree and tracking error is examined. Experimental results indicate that the observable degree of the target reflects tracking errors and can be also used to predict tracking errors.

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Appendix A. Computation process of each element of observation matrix

Each element of matrix $\mathbf{H}$ in Eq. (78) is calculated by

$$H(1,1) = \frac{\partial r}{\partial x} = \frac{x - x_t}{\sqrt{(x - x_t)^2 + (y - y_t)^2}} + \frac{x}{\sqrt{x^2 + y^2}} \quad (\text{A1})$$

$$H(1,2) = \frac{\partial r}{\partial v_x} = 0 \quad (\text{A2})$$

$$H(1,3) = \frac{\partial r}{\partial y} = \frac{y - y_t}{\sqrt{(x - x_t)^2 + (y - y_t)^2}} + \frac{y}{\sqrt{x^2 + y^2}} \quad (\text{A3})$$

$$H(1,4) = \frac{\partial r}{\partial v_y} = 0 \quad (\text{A4})$$

$$H(2,1) = \frac{\partial v_r}{\partial x} = \frac{v_y(y - y_t)(x - x_t) - v_x(y - y_t)^2}{[(x - x_t)^2 + (y - y_t)^2]^{\frac{3}{2}}} + \frac{xyv_y - y^2v_x}{(x^2 + y^2)^{\frac{3}{2}}} \quad (\text{A5})$$

$$H(2,2) = \frac{\partial v_r}{\partial v_x} = \frac{x_t - x}{\sqrt{(x - x_t)^2 + (y - y_t)^2}} - \frac{x}{\sqrt{x^2 + y^2}} \quad (\text{A6})$$

$$H(2,3) = \frac{\partial v_r}{\partial y} = \frac{v_x(x-x_t)(y-y_t) - v_y(x-x_t)^2}{[(x-x_t)^2 + (y-y_t)^2]^{\frac{3}{2}}} + \frac{xyv_x - x^2v_y}{(x^2 + y^2)^{\frac{3}{2}}} \quad (A7)$$

$$H(2,4) = \frac{\partial v_r}{\partial v_y} = \frac{y_t - y}{\sqrt{(x-x_t)^2 + (y-y_t)^2}} - \frac{y}{\sqrt{x^2 + y^2}} \quad (A8)$$

Appendix B. Computation process of each component of the observable degree function

The variables a_x , a_y , f_1 and f_2 in Eq. (79) can be written as

$$a_x = \frac{x_k}{\sqrt{x_k^2 + y_k^2}} + \frac{x_k - x_t}{\sqrt{(x_k - x_t)^2 + y_k^2}} \quad (B1)$$

$$a_y = \frac{y_k}{\sqrt{x_k^2 + y_k^2}} + \frac{y_k}{\sqrt{(x_k - x_t)^2 + y_k^2}} \quad (B2)$$

$$f_1 = (a_1 + a_2)(a_3a_4 + a_5a_6) - a_7(a_8a_9 + a_5a_6) + a_6(a_{10} + a_{11})(a_{12} + a_8) \quad (B3)$$

$$f_2 = a_4(a_1 + a_2)(a_{13} - a_5) + a_{14}(a_5a_6 + a_8a_9) - (a_{10} + a_{11})(a_8a_9 + a_6a_{13}) \quad (B4)$$

where

$$a_1 = \frac{-v_{y,k+1}x_{k+1}^2 + v_{x,k+1}x_{k+1}y_{k+1} - x_{k+1}^2y_{k+1} - y_{k+1}^3}{(x_{k+1}^2 + y_{k+1}^2)^{3/2}} \quad (B5)$$

$$a_2 = \frac{-v_{y,k+1}(x_{k+1} - x_t)^2 + v_{x,k+1}(x_{k+1} - x_t)y_{k+1} - (x_{k+1} - x_t)^2y_{k+1} - y_{k+1}^3}{[(x_{k+1} - x_t)^2 + y_{k+1}^2]^{3/2}} \quad (B6)$$

$$a_3 = \frac{v_{y,k}x_k^2 - v_{x,k}x_ky_k}{(x_k^2 + y_k^2)^{3/2}} + \frac{v_{y,k}(x_k - x_t)^2 - v_{x,k}(x_k - x_t)y_k}{[(x_k - x_t)^2 + y_k^2]^{3/2}} \quad (B7)$$

$$a_4 = \frac{x_{k+1}}{\sqrt{x_{k+1}^2 + y_{k+1}^2}} + \frac{x_{k+1} - x_t}{\sqrt{(x_{k+1} - x_t)^2 + y_{k+1}^2}} \quad (B8)$$

$$a_5 = -\frac{x_k}{\sqrt{x_k^2 + y_k^2}} + \frac{-x_k + x_t}{\sqrt{(x_k - x_t)^2 + y_k^2}} \quad (B9)$$

$$a_6 = \frac{y_{k+1}}{\sqrt{x_{k+1}^2 + y_{k+1}^2}} + \frac{y_{k+1}}{\sqrt{(x_{k+1} - x_t)^2 + y_{k+1}^2}} \quad (B10)$$

$$a_7 = \frac{-v_{y,k+1}x_{k+1}^2 + v_{x,k+1}x_{k+1}y_{k+1}}{(x_{k+1}^2 + y_{k+1}^2)^{3/2}} + \frac{-v_{y,k+1}(x_{k+1} - x_t)^2 + v_{x,k+1}(x_{k+1} - x_t)y_{k+1}}{[(x_{k+1} - x_t)^2 + y_{k+1}^2]^{3/2}} \quad (B11)$$

$$a_8 = \frac{y_k}{\sqrt{x_k^2 + y_k^2}} + \frac{y_k}{\sqrt{(x_k - x_t)^2 + y_k^2}} \quad (B12)$$

$$a_9 = \frac{x_{k+1}}{\sqrt{x_{k+1}^2 + y_{k+1}^2}} + \frac{x_{k+1} - x_t}{\sqrt{(x_{k+1} - x_t)^2 + y_{k+1}^2}} \quad (B13)$$

$$a_{10} = \frac{-x_{k+1}^3 + v_{y,k+1}x_{k+1}y_{k+1} - v_{x,k+1}y_{k+1}^2 - x_{k+1}y_{k+1}^2}{(x_{k+1}^2 + y_{k+1}^2)^{3/2}} \quad (B14)$$

$$a_{11} = \frac{-(x_{k+1} - x_t)^3 + v_{y,k+1}(x_{k+1} - x_t)y_{k+1} - v_{x,k+1}y_{k+1}^2 - (x_{k+1} - x_t)y_{k+1}^2}{[(x_{k+1} - x_t)^2 + y_{k+1}^2]^{3/2}} \quad (B15)$$

$$a_{12} = \frac{-v_{y,k}x_k^2 + v_{x,k}x_ky_k}{(x_k^2 + y_k^2)^{3/2}} + \frac{v_{x,k}x_k - v_{y,k}(x_k - x_t)^2 - x_t y_k}{[(x_k - x_t)^2 + y_k^2]^{3/2}} \quad (B16)$$

$$a_{13} = \frac{v_{y,k}x_ky_k - v_{x,k}y_k^2}{(x_k^2 + y_k^2)^{3/2}} + \frac{v_{y,k}(x_k - x_t)y_k - v_{x,k}y_k^2}{[(x_k - x_t)^2 + y_k^2]^{3/2}} \quad (B17)$$

$$a_{14} = \frac{v_{y,k+1}x_{k+1}y_{k+1} - v_{x,k+1}y_{k+1}^2}{(x_{k+1}^2 + y_{k+1}^2)^{3/2}} + \frac{v_{y,k+1}(x_{k+1} - x_t)y_{k+1} - v_{x,k+1}y_{k+1}^2}{[(x_{k+1} - x_t)^2 + y_{k+1}^2]^{3/2}} \quad (B18)$$

Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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